

A Consistent and Robust Test for Autocorrelated Jump Occurrences*

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Abstract

We develop a nonparametric test for the temporal dependence of jump occurrences in the population. The test is consistent against all pairwise serial dependence, and is robust to the jump activity level and the choice of sampling scheme. We establish asymptotic normality and local power property for a rich set of local alternatives, including both self-exciting and/or self-inhibitory jumps. Simulation study confirms the robustness of the test and reveals its competitive size and power performance over existing tests. In an empirical study on high-frequency stock returns, our procedure uncovers a wide array of autocorrelation profiles of jump occurrences for different stocks in different time periods.

Key words: autocorrelation, financial contagion, nonparametric test, self-excited jumps, self-inhibitory jumps

JEL classification: C12, C14, C58

1 Introduction

Jumps and diffusion are widely regarded as the key sources of variation in asset returns. Similar to stochastic volatility in diffusion, jumps are found to exhibit a rich variety of dynamic properties that are associated with different market environments. Negative jumps tend to occur more often and in clusters during turbulent periods in financial markets. At the intraday level, the jump dynamics may display seasonality over the course of a trading day (Khushanah, Chen and Hawkes 2018). A deep understanding of the dynamic nature of jumps is key to a comprehensive characterization of asset price formation over time. In particular, learning whether and how jumps are dependent over time has important

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implications on asset pricing (Bormetti et al., 2015) and derivative pricing (Ma, Pan, and Wang 2020), and is crucial for understanding the mechanism behind some market phenomena, for example, propagation of financial contagion (Aït-Sahalia, Cacho-Díaz, and Laeven 2015) and limit order arrivals (Bacry, Jaisson, and Muzy 2016).

The parametric modeling of the temporal dependence of jumps has been around for a few decades. Hawkes (1971) proposes one of the first parametric models for self-exciting jumps. The Hawkes model defines a parametric specification on the conditional jump intensity which is assumed to depend on the timing of past occurrences of jumps. Due to its intuitive and flexible parametric form, variants of Hawkes models have been applied to empirical finance; see Bacry, Mastromatteo, and Muzy (2015) and Hawkes (2018) for a survey. The popularity of the model lies in its ability to describe the jump-clustering phenomenon found in many financial time series.¹

While a parametric model offers analytical tractability and efficiency, it imposes constraints on the jump dynamics; in particular, only a certain kind of temporal dependence structure is allowed for a given kernel function. For example, a stationary Hawkes model with an exponential kernel implies that jumps must be self-exciting and that the associated autocorrelogram must decay exponentially (Hawkes 1971). The model-implied restrictions may rule out some empirically relevant jump dependence structure.² Without precise knowledge on the structure of jump dynamics, the modeler may pick a parametric specification for the kernel function out of analytical tractability or convenience. This may induce model misspecification risk, which could potentially lead to bias in inference.

Nonparametric inference of jump dynamics has become a vibrant area of research due to increased availability of high-frequency data and recent advance in methodology. While tests for the first-order properties of jumps are well studied,³ the inference problem for the second-order properties of jumps, in particular their serial dependence structure, is relatively new. To test for self-excitation of jumps, Boswijk, Laeven and Yang (2018; henceforth BLY) proposes a test that captures the co-movements of realized log-price process and spot jump intensity process estimated from realized jumps. Adopting in-fill asymptotics and a pathwise approach, consistency of the test requires an infinite number of jumps over a finite time window (i.e., infinite-activity jumps). By contrast, Corradi, Silvapulle, and Swanson (2018; henceforth CSS) tests the hypothesis of the existence of self-excited finite-activity jumps in the population, and delivers consistent inference under long-span asymptotics.

A number of issues arise from the use of existing tests. First, the two tests are powerful against certain departures from the null hypothesis, for example, BLY test is powerful in

- 1 For example, market returns during financial contagion and market downturn (Aït-Sahalia, Cacho-Díaz, and Laeven 2015), jumps in prices and volatilities (Maneesoonthorn, Forbes, and Martin 2017), and extremal returns (Embrechts, Liniger and Lin 2011; Chavez-Demoulin and McGill 2012).
- 2 Various kinds of jump dependence structure are documented in empirical quantitative finance. For example, limit order arrivals may display long-range persistence while obeying stationarity (Bacry et al. 2016). At the intraday level, self-depressing jumps (as opposed to self-exciting jumps) tend to occur in the opening session of a trading day (Khashanah, Chen and Hawkes 2018).
- 3 The literature on the inference of first-order properties of high-frequency jumps is vast; for example, see Aït-Sahalia and Jacod (2012) for a survey and Aït-Sahalia and Jacod (2014) for a textbook treatment.

the presence (or absence) of self-excited, infinite-activity jumps, while CSS displays power for lag-one autocorrelated finite-activity jumps. Another concern is on the robustness of test performance when the data are not fully compatible with the asymptotic theory and the underlying assumptions. For a finite dataset sampled at a certain frequency over a given horizon, it can be difficult to determine whether it fits better to a particular asymptotic scheme. As revealed by our simulation, the performance of existing tests can be unexpectedly sensitive to the choices of sampling frequency and sample horizon, sometimes yielding severely oversized or entirely powerless results. In terms of implementation, pre-testing and pre-estimation are necessary—CSS test requires a pre-testing stage for the presence of jumps, while BLY test requires a precise estimate of the degree of jump activities.

Motivated by the problems discussed above, we propose a set of procedures that deliver robust inference on the autocorrelation of jump occurrences for all possible degree of jump activity and for a wide range of empirically relevant sampling schemes. As our main contribution, we develop a consistent test under the proposed data generating process (DGP) with uncorrelated, stationary jump occurrences and arbitrary form of stochastic volatility and drift. The test is asymptotically valid under both in-fill and long-span asymptotics, and is powerful over a rich set of alternatives (e.g., self-exciting and self-inhibitory jumps, and jumps with cyclical autocorrelation profile). As confirmed by simulation, our test outperforms its competitors in terms of test power and robustness over sampling schemes and jump activity level. The superior performance is due to the distinct design of our test statistic, which is a weighted L^2 -norm of autocovariances of jump occurrences. The autocovariance is well-defined and can be estimated when jump occurrences are stationary.⁴ Furthermore, we examine the use of sample autocorrelogram as a visual tool to uncover the serial correlation structure of stationary jumps (e.g., persistence and seasonality). It allows for convenient pointwise inference up to some fixed lag order. Our procedures do not require pretesting for the presence of jumps and a priori knowledge of the jump activity level.

We confine ourselves in this article to a univariate series with strictly stationary jump process. While this assumption remains restrictive in some applications, the methodology developed under strict stationarity will be useful as the basis for extending the robust inference theory to the nonstationary setting.⁵ Stationarity is imposed on the unconditional joint distribution of the jump increment process, while conditional moments (e.g., conditional jump intensity) can be time-varying; so the condition is less restrictive than it appears. The methodology can be generalized to the inference of cross-correlation of jump occurrences.

The rest of the article is organized as follows. Section 2 sets out the DGP and introduces the autocovariance function as a measure of temporal dependence of jumps. Section 3 develops the nonparametric inference procedure and its asymptotic theory. Section 4 reports the simulation results and robustness analysis of the omnibus test. Section 5 presents the empirical application. Section 6 concludes. All proofs are given in the [Online Appendices](#).

4 The use of second-order measure (e.g., autocovariance) to describe the temporal dependence of stationary point processes has a long history (e.g., [Bartlett 1963](#); [Hawkes 1971](#); [Brillinger 1976](#)).

5 It is possible to extend the inference theory without the stationarity assumption. The time-varying mean intensity can be estimated using a nonparametric kernel estimator. The omnibus test can be applied to the *normalized* residual jump process, which is stationarity under the null hypothesis of uncorrelated jump occurrences.

2 Data Generating Process

Let X_t denote the log-price process defined in a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, P)$ which satisfies the usual conditions (Protter 2003). The following assumption specifies the DGP of X_t .

Assumption D: The log-price process X_t is a semi-martingale with decomposition

$$\begin{aligned} X_t &= X_0 + B_t + X_t^c + X_t^d \\ &= X_0 + \int_0^t b_s ds + \int_0^t \sigma_s dW_s + \left[\delta 1_{\{\|\delta\| \leq 1\}} * (\mu^X - \nu^X)_t + \delta 1_{\{\|\delta\| > 1\}} * \mu_t^X \right], \end{aligned}$$

where b_s and σ_s are locally bounded processes, $\delta = \delta(\omega, t, x)$ is some predictable function, μ^X is the jump measure, and ν^X is the compensator defined by $\nu^X(\omega, dt, dx) = \lambda_{t-}(\omega) F(dx) dt$, where $F(dx) = \frac{a^{(-)} 1_{\{x < 0\}} + a^{(+)} 1_{\{x > 0\}}}{|x|^{1+\beta}} dx$ for $0 < \beta < 2$ and $a^{(-)}, a^{(+)} > 0$, and λ_t is locally bounded and mean-stationary with $E(\lambda_t) \equiv \bar{\lambda} < \infty$.

Assumption D places restrictions on the jump process X_t^d : The jump process X_t^d is infinitely active ($\beta > 0$), and the jump increments $dX_t^d := X_t^d - X_{t-}^d$ form a strictly stationary process (as implied by the time-invariant Lévy measure $F(dx)$ and the stationarity of λ_t). The DGP is similar to those found in the literature; for example, Aït-Sahalia and Jacod (2009a,b), Jing et al. (2012), and BLY (2018).⁶ Autocorrelation of jumps may be induced through $\lambda_t(\omega)$, which is viewed as the stochastic scaling factor.

Our goal is to characterize how the occurrences and non-occurrences of jumps are temporally dependent. To this end, we introduce the Bernoulli process

$$I_t(\Delta, \epsilon; \omega) := 1\{|X_t^d - X_{t-\Delta}^d| > \epsilon\}, \quad \Delta > 0, \epsilon > 0, \text{ and } \omega \in \Omega_J,$$

where $\Omega_J := \{\omega : |X_t^d - X_{t-\Delta}^d| > 0 \text{ for some } t > 0\}$, that is, the set of all sample paths on which jumps occurs, and $1\{S\}$ is the indicator function which equals one if S occurs, and zero otherwise. The Bernoulli process retains only information about the timing of those jumps occurring in $(t - \Delta, t]$ with magnitude exceeding the threshold ϵ . Information regarding the jump size is removed (except for whether it exceeds ϵ or not).

The stationarity condition in Assumption D implies that all unconditional moments of jump increments are time-invariant if they exist. In particular, due to boundedness of the Bernoulli process, all moments of $I_t(\Delta, \epsilon)$ are well defined and remain constant over time. We define the *population autocovariance function* of $I_t(\Delta, \epsilon)$ as follows:

$$\gamma(s; \Delta, \epsilon) := \text{Cov}_{\Omega_J}\{I_t(\Delta, \epsilon), I_{t+s}(\Delta, \epsilon)\}.$$

It fully captures all pairwise temporal dependence of $I_t(\Delta, \epsilon)$ conditional on the occurrence of jumps.⁷

- 6 They consider a slightly more general jump process with a generic truncation function and an additional singular element in the Lévy measure.
- 7 Indeed, for $s \neq t$, the joint distribution of $I_s(\Delta, \epsilon)$ and $I_t(\Delta, \epsilon)$ is fully characterized by their first two moments (mean, variance, and covariance). This follows from the binary nature of the jump occurrence process.

Under Assumption D, the moments of $I_t(\Delta, \epsilon)$ (e.g., $E[I_t(\Delta, \epsilon; \omega)]$ and $\gamma(s; \Delta, \epsilon)$) are well defined for sufficiently large ϵ . This is illustrated in the following examples.

Example 1 Suppose the DGP is given by Assumption D, and $\lambda_t = \lambda_\infty + \int_0^t \varphi(t-s) dN_s^\epsilon$, where $N_t^\epsilon := \sum_{0 < s \leq t} 1\{|dX_s^d| > \epsilon\}$ for some fixed $\epsilon > \epsilon_0 > 0$. Consider the exponential kernel $\varphi(x) = \alpha e^{-\zeta x} 1_{\{x > 0\}}$ for $0 < \alpha < \zeta$. Provided that $\epsilon > \epsilon_0 := \left[\frac{2(a^{(-)} + a^{(+)})}{\zeta \beta} \right]^{1/\beta}$, the autocovariance function is well defined and is given by $\gamma(s; \Delta, \epsilon) = A_\epsilon e^{-B_\epsilon |s|} \Delta^2 + o(\Delta^2)$ as $\Delta \downarrow 0$ for $s > 0$. The parameters A_ϵ, B_ϵ are strictly positive and depend on the threshold ϵ and model parameters (see the [Online Appendix](#) for details). Note that the condition $0 < \alpha < \zeta$ ensures the stationarity of λ_t , and the lower bound ϵ_0 guarantees $\bar{\lambda} < \infty$. While the jump process X_s^d remains infinitely active and has infinite jump intensity, those jump increments dX_s^d with magnitude greater than ϵ display self-excitation via λ_t .

Example 2 Suppose the DGP in Example 1 except that a periodic kernel is used: $\varphi(x) = \cos\left(\frac{2\pi x}{\tau}\right) 1_{\{0 < x < m\tau\}}$, where $\tau > 0$ and m is a positive integer. Then the jump process is stationary and generates jumps that exhibit both self-exciting and inhibitory behaviors over different lag lengths. The autocovariance function $\gamma(s; \Delta, \epsilon)$ alternates in sign and decays in magnitude with s , although its analytical form is unknown. More generally, for any kernel function that satisfies $\int_0^\infty \varphi(x) dx < 1$, the jump process is stationary, and the autocovariance function is the solution of a Volterra integral equation (see [Hawkes 1971](#), Equation (11)).

The population autocovariance function $\gamma(s; \Delta, \epsilon)$ is unknown in practice because the jump process X_t^d is unobserved; nonetheless, under suitable asymptotics, it is possible to separate jump from diffusive increments in a time-series sample of log-prices $\{X_0, X_1, \dots, X_n\}$. Let $[0, T]$ be the sample horizon, and let $\Delta_i^n X := X_i - X_{i-1}$ denote the discretized sequence of log-returns over an interval of length Δ , so that $\Delta = \frac{T}{n}$. To separate the jumps from the diffusive changes, we construct the *filtered jump sequence* $\left\{ g_k\left(\frac{\Delta_i^n X}{u_n}\right) \right\}_i$, where $g_k\left(\frac{x}{u}\right) := \min(1, |x/u|^k)$ for $k > 0$ and $g_\infty\left(\frac{x}{u}\right) := 1\{|x| > u\}$ ([Aït-Sahalia and Jacod 2009a, 2009b](#); [Jing et al. 2012](#)). By setting the sequence u_n and the power k judiciously, it is possible to disentangle jumps from diffusive increments asymptotically.⁸

The following proposition motivates the choice of tuning parameter k in the filtered jump sequence. To set the stage, we assume that the sequence u_n takes the form $u_n = c\Delta^\sigma$. For a given $\Delta > 0$, denote $\Delta X_t = X_t - X_{t-\Delta}$. Define the *demeaned* filtered jump $\tilde{g}_k\left(\frac{\Delta X_t}{c\Delta^\sigma}\right) := g_k\left(\frac{\Delta X_t}{c\Delta^\sigma}\right) - E\left[g_k\left(\frac{\Delta X_t}{c\Delta^\sigma}\right)\right]$, which is interpretable as jump surprise beyond the average jump intensity. For any two expressions A and B , we write $A \simeq B$ if $\frac{A}{B} = 1 + o_p(1)$ in the limit.

8 The filtered sequence records as one ($|\Delta_i^n X/u_n|^k$) when the discretized change $\Delta_i^n X$ is larger (smaller) than u_n in magnitude. Setting a larger k and u_n discounts smaller increments more heavily, but has a smaller risk of misclassifying a diffusive increment as a jump.

Proposition 1 Under Assumption D, the following statements hold as $\Delta \downarrow 0$.

i. For $0 < k < \frac{1-\beta\varpi}{1-2\varpi}$,

$$\tilde{g}_k\left(\frac{\Delta X_t}{c\Delta^\varpi}\right) \simeq \left|\frac{\sigma_t \Delta W_t}{c\Delta^\varpi}\right|^k 1_{\{|\Delta X_t| \leq c\Delta^\varpi\}} = O_p(\Delta^{k(1/2-\varpi)}).$$

ii. For $k = \frac{1-\beta\varpi}{1-2\varpi}$,

$$\tilde{g}_k\left(\frac{\Delta X_t}{c\Delta^\varpi}\right) \simeq \left|\frac{\sigma_t \Delta W_t}{c\Delta^\varpi}\right|^k 1_{\{|\Delta X_t| \leq c\Delta^\varpi\}} + 1_{\{|\Delta X_t^d| > c\Delta^\varpi\}} - P(|\Delta X_t^d| > c\Delta^\varpi) = O_p(\Delta^{(1-\beta\varpi)/2}).$$

iii. For $k > \frac{1-\beta\varpi}{1-2\varpi}$,

$$\tilde{g}_k\left(\frac{\Delta X_t}{c\Delta^\varpi}\right) \simeq 1_{\{|\Delta X_t^d| > c\Delta^\varpi\}} - P(|\Delta X_t^d| > c\Delta^\varpi) = O_p(\Delta^{(1-\beta\varpi)/2}).$$

The result is not entirely trivial. It indicates that the demeaned filtered jump process can remove the diffusive increments and retain the infinitely active jumps in the limit (regardless of whether T is fixed or diverges to ∞), provided that a sufficient large power k is chosen. We note that the minimal k that enables separation between diffusive and jump variation is smaller compared with BLY (2018). The difference reflects the tradeoff between moment restrictions (via k) and serial dependence structure (stationarity of jump increments). The use of $\tilde{g}_k\left(\frac{\Delta X_t}{c\Delta^\varpi}\right)$ with a smaller k will retain more jumps and hence lead to more powerful tests (see Remarks 1 and 3). This is desirable when jumps do not occur very frequently in the sample but stationarity is a reasonable assumption to impose on the jump process.

Motivated by the above proposition, we impose the following assumption on k and u_n .

Assumption G: The sequence $g_k\left(\frac{\Delta_t^n X}{u_n}\right)$ is chosen with $k > \frac{1-\beta\varpi}{1-2\varpi}$ and $u_n = c\Delta^\varpi$, where $c > 0$ and $0 < \varpi < \frac{1}{2}$.

Remark 1 BLY (2018) requires the stronger condition $k > \frac{1-\beta\varpi}{1/2-\varpi}$. Without a priori knowledge of the jump activity level β , our condition on k can be tightened to $k > \frac{1}{1-2\varpi} = \sup_{\beta \in (0,2)} \frac{1-\beta\varpi}{1-2\varpi}$. This is due to the use of demeaned sequence $\tilde{g}_k\left(\frac{\Delta_t^n X}{c\Delta^\varpi}\right)$ as opposed to the original filtered jump sequence $g_k\left(\frac{\Delta_t^n X}{c\Delta^\varpi}\right)$.

3 Inference of Autocorrelations of Jump Occurrences

In this section, we develop a joint test for autocorrelations of jump occurrences under weak assumptions on the sampling scheme and jump processes. The test statistic makes use of the filtered sequence as inputs. The test delivers valid size even if no jumps are present in the sample series.⁹

9 Both methods would conclude that jump occurrences have zero autocorrelations if all time periods contain big jumps (i.e., the filtered jump series $g_{n,k,i}$ is identically one). This problem is likely caused by having an excessively small cutoff u_n and is readily verifiable by the user.

3.1 Sample Autocorrelogram of Jump Occurrences

We first introduce the sample autocovariance measure for jump occurrences. For notational convenience, we let $g_{n,k,i} := g_k\left(\frac{\Delta_n^i X}{u_n}\right)$ and $\bar{g}_{n,k} := \frac{1}{n} \sum_{i=1}^n g_{n,k,i}$. Define the *sample autocovariance function* of $g_{n,k,i}$:

$$\hat{\gamma}_{n,k}(\ell) := \frac{1}{n} \sum_{i=1}^{n-\ell} [g_{n,k,i} - \bar{g}_{n,k}][g_{n,k,i+\ell} - \bar{g}_{n,k}]. \quad (1)$$

Our goal is to construct a joint test that is consistent and powerful against autocorrelated jump occurrences over *all* lag orders. The null hypothesis is given by Assumption D₀, stated formally below.

Assumption D₀: Assumption D holds. In addition, the null hypothesis holds:

$$H_0 : \gamma(s; \Delta, \epsilon) = 0 \text{ for all } s \neq 0 \text{ and all } \Delta, \epsilon > 0.$$

Define the test statistic

$$L(k) := n \sum_{\ell=1}^{n-1} w_b(\ell) \hat{\gamma}_{n,k}^2(\ell),$$

where $w_b(\ell)$ specifies the weight on squared lag- ℓ autocovariance. For notational simplicity, we suppress the dependence of the test statistic on n and h .

Intuitively, the test statistic $L(k)$ summarizes all information regarding the temporal dependence of all pairs of jump occurrences by means of a weighted L^2 -norm on $\{\hat{\gamma}_{n,k}(\ell)\}_{\ell>0}$. The weight decreases with lag, reflecting the economic intuition that jumps closer to each other in occurrence time has stronger association. The use of L^2 -norm avoids possible cancellation between positive and negative autocovariances due to self-excited and self-inhibitory jumps. The test reduces to the portmanteau test of Box and Pierce (1970) if the weights are all equal and the sum is truncated at a finite lag order; as such, $L(k)$ test can be viewed as a generalized Box–Pierce test (Hong 1999). Since the weighted sum spans all lags in the limit, the test is consistent over all possible lag orders.

For valid large-sample inference, the weight function $w(\cdot)$, the tuning parameter h (which controls the decay rate of the weights on squared autocovariances), and the sampling interval $\Delta = \frac{T}{n}$ need to obey certain restrictions. Let $\Delta = \Delta_n = O(n^{-\nu})$ and $h = O(n^\xi)$ as $n \rightarrow \infty$. The following assumption accommodates both long-span ($0 < \nu < 1$; expanding T and shrinking Δ) and fixed-span ($\nu = 1$; fixed T and shrinking Δ) asymptotics, depending on the jump activity level β and the user-specified cut-off parameter ϖ ($0 < \varpi < 0.5$ by Assumption G).

Assumption T_L: (i) $w_b(\ell) = \frac{1}{b} w\left(\frac{\ell}{b}\right)$, where $w(x)$ is defined on $\mathbb{R}_+$ and satisfies $\int_0^\infty w(x)dx = 1$ and $\mathcal{W}_2 = \int_0^\infty w^2(x)dx < \infty$; (ii) $\nu \in (0, 1] \cap (0, \frac{1}{1-\beta\varpi})$ and $\xi \in (\max[0, 2\nu(1 - \beta\varpi) - 1], 1)$.

Remark 2 Assumption T_L(i) defines the set of valid weight functions that are typical in nonparametric kernel estimation. Assumption T_L(ii) specifies the admissible sampling schemes (controlled by ν) and penalties on autocovariances as a function of lag (controlled by ξ) such that $L(k)$ test is consistent. The restriction $\nu < \frac{1}{1-\beta\varpi}$ ensures that the set of admissible (ν, ξ) is non-empty. The set of admissible (ν, ξ) becomes larger as β increases

(see Figure 1(b) and (c)). The general rule is that more lags are needed (i.e., larger ξ , hence larger h) when the jump activity level is lower (smaller β), the trimming of small jumps is more aggressive (smaller ϖ), or when the sampling intervals shrink more quickly (bigger ν). The restriction $\nu > 0$ requires $\Delta \downarrow 0$ in the limit. Such in-fill asymptotics is necessary to ensure (i) the consistency of $L(k)$ test under the null hypothesis (Assumption D_0), and (ii) that variation due to drift (of order $O_p(\Delta)$) is dominated by diffusive or jump variation (of order $O_p(\sqrt{\Delta})$) as $\Delta \downarrow 0$,

The following theorem establishes asymptotic normality of the studentized statistic $t_{L(k)}$ under the null hypothesis (the subscript 0 in $\widehat{E}_0[\cdot]$ and $\widehat{\text{Var}}_0[\cdot]$ indicates that the mean and variance are evaluated under H_0).

Theorem 1 Suppose Assumptions D_0 , G , and T_L hold. Then for all $r > 0$ and $p > 1$, we have

$$t_{L(k)} := \frac{L(k) - \widehat{E}_0[L(k)]}{\sqrt{\widehat{\text{Var}}_0[L(k)]}} \xrightarrow{d} N(0, 1),$$

as $n \rightarrow \infty$, where

$$\begin{aligned}\widehat{E}_0[L(k)] &= [\widehat{\text{Var}}_0(g_{n,k,i})]^2 \sum_{\ell=1}^{n-1} \left(1 - \frac{\ell}{n}\right) w_b(\ell), \\ \widehat{\text{Var}}_0[L(k)] &= \frac{1}{n} [\widehat{\text{Var}}_0(g_{n,2k,i}) + r\Delta^p]^2 \sum_{\ell=1}^{n-1} \left(1 - \frac{\ell}{n}\right) w_b^2(\ell) \\ &\quad + 2[\widehat{\text{Var}}_0(g_{n,k,i}) + r\Delta^p]^4 \sum_{\ell=1}^{n-1} \left(1 - \frac{\ell}{n}\right) \left(1 - \frac{\ell+1}{n}\right) w_b^2(\ell), \\ \widehat{\text{Var}}_0(g_{n,k,i}) &= \frac{1}{n-1} \sum_{i=1}^n (g_{n,k,i} - \bar{g}_{n,k})^2, \quad \bar{g}_{n,k} = \frac{1}{n} \sum_{i=1}^n g_{n,k,i}.\end{aligned}$$

Remark 3 The BLY test detects self-excitation of jumps by taking a pathwise approach. By contrast, $L(k)$ test places restrictions on population attributes (e.g., strict stationarity on jump increments; see Assumption D); in this sense, $L(k)$ test is similar to the CSS test. Another key distinction lies in the way temporal dependence of jumps is measured: BLY (2018) studies the co-jumps of the observed log-return series and the estimated spot intensity process, whereas $L(k)$ test directly captures the lead-lag relationship of jump occurrences using the autocovariance measure. The two test statistics summarize the information regarding jump dependence in different ways. Furthermore, compared with BLY test, the cost borne by $L(k)$ test for requiring stationarity of jump increments is offset by an increase of test power due to less restrictive conditions on the power k (Assumption G).¹⁰ We expect that weaker conditions on k induce more variation due to smaller jumps and thus enhances

10 For infinite-activity jumps, the BLY test requires that the power be greater than $\frac{1-\varpi\beta}{1/2-\varpi}$ (BLY 2018, Theorem 2), which is twice as high as the minimal k allowed for $L(k)$ test (Assumption G). This is due to the difference in the stochastic orders of $g_{n,k,i}$ ($= O_p(\Delta^{1-\varpi\beta})$), which enters the BLY test statistic, and its demeaned form $\tilde{g}_{n,k,i}$ ($= O_p(\Delta^{(1-\varpi\beta)/2})$; see Proposition 1), which enters $L(k)$. The latter is well defined under Assumption D .

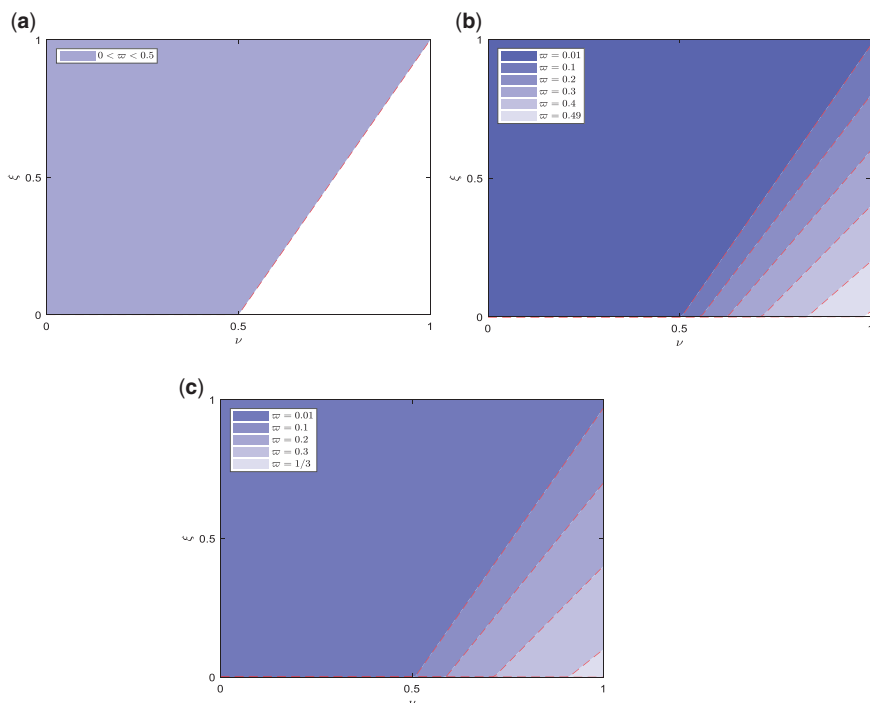


Figure 1. Admissible values of (ν, ξ) for different jump activity levels β : (a) $\beta = 0$; (b) $\beta = 1$; (c) $\beta = 1.5$.

the power of $L(k)$ test. This will be seen in the simulation study in Section 4. Further comparison between BLY and $L(k)$ tests is detailed in the [Online Appendix](#).

Remark 4 The variance estimator $\widehat{\text{Var}}_0[L(k)]$ is composed of two terms: the first term dominates the second one when $\frac{1}{2(1-\beta m)} < \nu \leq 1$, whereas the second term dominates the first one when $0 < \nu \leq \frac{1}{2(1-\beta m)}$. The presence of both terms guarantees robustness of the test to different asymptotic schemes. The inclusion of $r\Delta^p$ ensures that $L(k)$ test is implementable when jumps are absent (as otherwise the denominator of $t_{L(k)}$ would be zero), thereby avoiding the need to pre-test for jumps. It does not affect the asymptotic properties of the test.¹¹ In our simulation and empirical studies, we will set $r = 1$ and $p = 2$.

3.2 Test Power under Local Alternatives

We now study the power of $L(k)$ test by considering local alternatives in which jumps are autocorrelated. The local alternatives and the necessary higher moment restrictions are given in Assumption D_a .

Assumption D_a : Assumption D holds. In addition, there exists a constant $K > 0$ such that the following conditions are satisfied for all n .

- 11 For any $r > 0$ and $p > 1$, the extra term $r\Delta^p$ is dominated by $\text{Var}_0(g_{n,k,i})$, which is $O(\Delta^{1-\beta m})$ for all k satisfying Assumption G (see Lemmas 2 and 3 in the [Online Appendix](#)).

- i. $\gamma(\ell\Delta) = f(\frac{\ell}{b}) \frac{\Delta^{1-\beta\varpi}}{n^{1/2}b^{\delta/2}}$ for $0 < \nu \leq \frac{1}{2(1-\beta\varpi)}$, and $\gamma(\ell\Delta) = f(\frac{\ell}{b}) \frac{\Delta^{(1-\beta\varpi)/2}}{n^{3/4}b^{\delta/2}}$ for $\frac{1}{2(1-\beta\varpi)} < \nu \leq 1$, where $0 \leq \delta \leq \frac{1}{2}$ and $f(x)$ is locally bounded around $x = 0$ satisfying $0 < \tilde{K} := \int_0^\infty w(x)f^2(x)dx < \infty$.
- ii. $\sum \sum \sum_{j_1, j_2, j_3=1}^n |\text{cum}(\tilde{g}_i, \tilde{g}_{j_1}, \tilde{g}_{j_2}, \tilde{g}_{j_3})| \leq K\Delta^{4(1-\beta\varpi)}$ for all $i = 1, \dots, n$.
- iii. $\sum_{\ell \neq 0} w^2(\ell) \text{Var}(\tilde{g}_i^2 \tilde{g}_{i+\ell}^2) \leq K\Delta^{2(1-\beta\varpi)}$ for all $i = 1, \dots, n$, and $\sum_{\ell \neq 0} w^2(\ell) \text{Var}(\tilde{g}_i \tilde{g}_{i+\ell} \tilde{g}_j \tilde{g}_{j+\ell}) \leq K\Delta^{4(1-\beta\varpi)}$ for all $i, j = 1, \dots, n$.
- iv. $\sum \sum_{(i_1, j_1) \neq (i_2, j_2)} |\text{Cov}(\tilde{g}_{i_1}^2 \tilde{g}_{j_1}^2, \tilde{g}_{i_2}^2 \tilde{g}_{j_2}^2)| \leq K\Delta^{2(1-\beta\varpi)}$ for all $i_1, j_1 = 1, \dots, n$, and $\sum \sum \sum_{(i_1, j_1, p_1, q_1) \neq (i_2, j_2, p_2, q_2)} |\text{Cov}(\tilde{g}_{i_1} \tilde{g}_{j_1} \tilde{g}_{p_1} \tilde{g}_{q_1}, \tilde{g}_{i_2} \tilde{g}_{j_2} \tilde{g}_{p_2} \tilde{g}_{q_2})| \leq K\Delta^{4(1-\beta\varpi)}$ for all $i_1, j_1, p_1, q_1 = 1, \dots, n$.

Remark 5 Condition (i) defines precisely the set of local alternatives against which $L(k)$ test exhibits power.¹² We expect that $L(k)$ test is powerful for alternatives in which $\gamma(\ell\Delta)$ alternates in signs as lag order varies; for example, periodic functions such as $f(x) = \sin(\frac{2\pi x}{\tau})$ clearly satisfy condition (i). For fixed lag order ℓ , as the sample size n increases $\gamma(\ell\Delta)$ diminishes to zero at a certain rate that is determined by how quickly the variance of $L(k)$ vanishes to zero. This diminishing rate of $\gamma(\ell\Delta)$ determines the power of $L(k)$ test. The decay rate of the weighting function (controlled by ξ) plays a key role in affecting the test power.¹³ Condition (ii) is similar to Assumption $D_a(\ell)$ (iii) but somewhat stronger as it places restrictions on the dependence of all quadruples of $\tilde{g}_i$. Conditions (iii) and (iv) control the serial dependence of $\tilde{g}_i$ up to the eighth order.

The asymptotic power result of $L(k)$ test is stated in the following theorem.

Theorem 2 Suppose Assumptions D_a , G , and T_L hold. Then the following statements are true.

- a. For $0 \leq \delta < \frac{1}{2}$,

$$P[|t_{L(k)}| > c] \rightarrow 1$$
as $n \rightarrow \infty$, where c is any positive constant.

- a. For $\delta = \frac{1}{2}$,

$$t_{L(k)} = \theta_k + X + o_p(1),$$
as $n \rightarrow \infty$, where X is a zero-mean $O_p(1)$ random variable,

$$\theta_k := \begin{cases} \frac{\tilde{K}}{\sqrt{2W_2}\sigma_k^4} & \text{for } 0 < \nu \leq \frac{1}{2(1-\beta\varpi)}, \\ \frac{\tilde{K}}{\sqrt{W_2}\sigma_{2k}^2} & \text{for } \frac{1}{2(1-\beta\varpi)} < \nu \leq 1, \end{cases}$$

and $\sigma_k^2 := \lim_{n \rightarrow \infty} \frac{\text{Var}_0(g_{n,k,i})}{\Delta^{1-\beta\varpi}}$.

- 12 The local boundedness of f and that $\tilde{K} < \infty$ imply the uniform boundedness of f ; hence, $|\gamma(s)| \rightarrow 0$ as $n \rightarrow \infty$ for all $s \neq 0$.
- 13 Comparing the local alternatives in Assumption $D_a(\ell)$ with Assumption $D_a(\ell)(\hat{\ell})$, we see that $\gamma(\ell\Delta)$ diminishes at a faster rate than that of the pointwise $\hat{\rho}_{n,k}(\ell)$ test. Given the nature of $L(k)$ test which aggregates more information of jump autocorrelation over multiple lags, we expect that $L(k)$ test is powerful against more directions than $\hat{\rho}_{n,k}(\ell)$ test for given ℓ .

3.3 The Case for Finite-Activity Jumps

The analysis of finite-activity jumps has a long history in many empirical applications, and is still quite prevalent nowadays for the inference of jump contagion using lower-frequency financial data (e.g., Ait-Sahalia, Cacho-Diaz, and Laeven 2015). Does the proposed non-parametric test apply in this case? To address this question, let us consider the following jump diffusion model in which the jumps are finitely active. It includes Hawkes and Poisson processes as special cases.

Assumption D': The log-price process X_t is a semi-martingale with decomposition

$$\begin{aligned} X_t &= X_0 + B_t + X_t^c + X_t^d \\ &= X_0 + \int_0^t b_s ds + \int_0^t \sigma_s dW_s + \int_0^t J_s dN_s, \end{aligned}$$

where b_s and σ_s are locally bounded processes, N_t is the counting process of a stationary simple point process with conditional intensity λ_t ,¹⁴ and J_t is a strictly stationary process with distribution F_J and is independent of $\mathcal{F}_{t-}$.

Although the DGP shares the same semi-martingale decomposition as in Assumption D, the specification of the jump process X_t^d is distinctively different. This paradigm shift is necessary in order to match the specifications commonly adopted in the empirical modeling of both the infinite- and finite-activity jumps. The scaling of jumps is done via J_t , while the temporal dynamics of the occurrence of jumps is controlled by λ_t , which now plays the role of the stochastic intensity process of the counting process N_t . The clash of the notation λ_t is intentional as the autocorrelation of jumps, if any, is made possible via λ_t in both cases.

The following example is analogous to Example 1 under infinite-activity jumps.

Example 3. Suppose the DGP is given by Assumption D' where $\lambda_t = \lambda_\infty + \int_0^t \varphi(t-s)N_s ds$ and where $N_t := \sum_{0 < s \leq t} 1\{|dX_s^d| > 0\}$ is the counting process for jumps. Consider the exponential kernel $\varphi(x) = \alpha e^{-\zeta x} 1_{\{x > 0\}}$ ($0 < \alpha < \zeta$). Then λ_t is stationary. Jumps are finitely active with unconditional intensity $\bar{\lambda} = \frac{\lambda_\infty}{1-\alpha}$. The autocorrelation of jump occurrences is given by (Hawkes 1971)

$$\gamma(s; \Delta, \epsilon) = \frac{\alpha \bar{\lambda} (2\zeta - \alpha)}{2(\zeta - \alpha)} e^{-(\zeta - \alpha)s} \Delta^2 + o(\Delta^2) \text{ for } s > 0. \quad (2)$$

Consequently, jumps are self-exciting (due to positivity of γ) and decays exponentially with lag.

Unlike infinite-activity jump processes, it is well known that a finite-activity jump process generates only finitely many jumps in a finite time window. This distinctive feature calls for a different asymptotic set-up. Importantly, fixed-span asymptotics is no longer valid for statistical inference as there are at most finitely many jumps to be sampled over a fixed horizon.

14 For small $\Delta > 0$, the conditional law of N_t is characterized by $P(\Delta N_t = 0 | \mathcal{F}_{t-}) = 1 - \lambda_t \Delta + o(\Delta)$, $P(\Delta N_t = 1 | \mathcal{F}_{t-}) = \lambda_t \Delta + o(\Delta)$, and $P(\Delta N_t > 1 | \mathcal{F}_{t-}) = o(\Delta)$, where λ_t is the conditional intensity defined by $\lambda_t := \frac{P(\Delta N_t \geq 1 | \mathcal{F}_{t-})}{\Delta}$. The unconditional intensity $\bar{\lambda} := E(\lambda_t)$ is a finite constant by stationarity.

The only available option to ensure test consistency is to allow the sample horizon to expand in the limit. Fortunately, the analytical framework we established earlier remains applicable under Assumption D'. By letting $\beta = 0$, it is clear from Assumption $T_L(ii)$ that fixed-span asymptotics ($\nu = 1$) is ruled out, and the admissible range of ν is $0 \leq \nu < 1$ (see Figure 1(a)). The bottom line is that T has to grow with n , as otherwise the intervals would shrink too quickly and there would be insufficient jumps available for consistent autocovariance estimation.¹⁵

The consistency and local power property of $L(k)$ test when jumps are of finite-activity are preserved. This is stated in the following corollary.

Corollary 1: *The conclusions in Theorems 1 and 2 remain valid if Assumption D is replaced by Assumption D' and β is set to 0.*

3.4 Sample Autocorrelogram of Jump Occurrences

As a by-product of our theory, the sample autocovariance function (1) readily lends itself to visual analysis after suitable normalization. We introduce the *lag- ℓ sample autocorrelation function*:

$$\hat{\rho}_{n,k}(\ell) := \frac{\hat{\gamma}_{n,k}(\ell)}{\hat{\gamma}_{n,k}(0) + r\Delta^p},$$

where $r > 0$ and $p > 1$. The term $r\Delta^p$ plays a similar role as that in the variance of $L(k)$ (see Remark 6): it ensures that $\hat{\rho}_{n,k}(\ell)$ remains well-defined when the sample process exhibits no jumps (in which case $\hat{\gamma}_{n,k}(0)$ and $\hat{\gamma}_{n,k}(\ell)$ are both zero but $r\Delta^p$ is strictly positive), while its presence does not asymptotically alter the interpretation of $\hat{\rho}_{n,k}(\ell)$ as the sample autocorrelation of jump occurrences (as $r\Delta^p$ is strictly dominated by $\hat{\gamma}_{n,k}(0)$ asymptotically); consequently, the value of $\hat{\rho}_{n,k}(\ell)$ lies in $(-1, 1)$.

The following assumption imposes mild restrictions on the sampling scheme.

Assumption T_γ : $\Delta = O(n^{-\nu})$ as $n \rightarrow \infty$, where $\nu \in (0, 1] \cap (0, \frac{1}{2(1-\beta m)})$.

The restriction ensures consistency of the sample autocorrelogram for fixed lag order. Compared with the requirements for $L(k)$ test (Assumption T_L), Assumption T_γ prohibits the sampling intervals from shrinking too quickly when jumps are of finite activities ($\beta = 0$); see Remark 4; in particular, the assumption rules out fixed-span asymptotics ($\nu = 1$).

We now establish the asymptotic normality of $\hat{\rho}_{n,k}(\ell)$ under the null hypothesis of uncorrelated jump occurrences.

Theorem 3 *Suppose Assumptions D_0 , G , and T_γ hold. Then, for any fixed $\ell > 0$, we have*

$$\sqrt{n}\hat{\rho}_{n,k}(\ell) \xrightarrow{d} N(0, 1)$$

as $n \rightarrow \infty$.

The above result delivers a convenient large-sample test for the autocorrelations of jump occurrences over a finite range of lags $\ell = 1, \dots, M$, where M is some finite maximum lag order prespecified by the user. The test can be visually implemented by interlacing the

15 Recall that a simple point process has at most a finite number of jumps in a finite time interval.

sample autocorrelogram plot (i.e., the plot of $\hat{\rho}_{n,k}(\ell)$ against $\ell = 1, \dots, M$) with a pair of parallel lines at levels $\mp z_{\alpha/2}/\sqrt{n}$ representing the critical band ($z_{\alpha/2}$ denotes the standard normal variate associated with a tail probability of $\alpha/2$). Note that the critical values do not depend on the lag order ℓ . We conclude that the lag- ℓ autocorrelation is significant at the level α if $\hat{\rho}_{n,k}(\ell)$ goes beyond the critical band.

Next we examine the power property of the autocorrelation test based on $\hat{\rho}_{n,k}(\ell)$ under a suitable sequence of alternatives. To this end, we impose more explicit structure on the temporal dependence of jump occurrences. The alternatives and restrictions on the temporal dependence structure are given in the following assumption. To simplify notation, we denote $\tilde{g}_i = \tilde{g}_{n,k,i} := g_{n,k,i} - E(g_{n,k,i})$.

Assumption D_a(ℓ): Assumption D holds. In addition, there exists a constant $K > 0$ such that the following conditions are satisfied for all n :

- i. $\gamma(\ell\Delta) = \frac{K_\ell}{n^\delta}$, where $0 \leq \delta \leq \frac{1}{2}$ and $K_\ell \neq 0$ with $\sum_{\ell>0} |K_\ell| \leq K$.
- ii. $\sum_{j=1, j \neq i}^n |\text{Cov}(\tilde{g}_i^2, \tilde{g}_j^2)| \leq K\Delta^{2(1-\beta\varpi)}$ for all $i = 1, \dots, n$.
- iii. $\sum_{j=1}^n |\text{Cov}(\tilde{g}_i \tilde{g}_{i+\ell}, \tilde{g}_j \tilde{g}_{j+\ell})| \leq K\Delta^{2(1-\beta\varpi)}$ for all $i = 1, \dots, n$.

Remark 6 Condition (i) sets out a sequence of DGPs in which jump occurrences are autocorrelated at lag order ℓ (fixed). The autocovariance function $\gamma(\ell\Delta)$ may remain constant ($\delta=0$) or diminish to zero ($\delta>0$) as the sample size n increases, and is assumed to satisfy absolute summability over lags (as implied by $\sum_{\ell>0} |K_\ell| \leq K$). The condition encompasses finite-activity jump processes with finite covariance density (Bartlett 1963). In particular, if $K_\ell = Ae^{-(\zeta-\alpha)|\ell|\Delta} \Delta^2$, where $0 < \alpha < \zeta$ and $A > 0$ (depending on α and ζ), the DGP represents a stationary Hawkes process with exponential kernel $\varphi(x) = \alpha e^{-\zeta x}$ (Hawkes 1971). The condition also accommodates infinite-activity jump processes with autocorrelated jumps. Conditions (ii) and (iii) impose restrictions on the serial dependence of $\tilde{g}_i$ up to the fourth order.

We expect that the autocorrelogram test based on $\hat{\rho}_{n,k}(\ell)$ for fixed lag order $\ell \neq 0$ is powerful against autocorrelated jump occurrences that are separated apart by $\ell\Delta$. This is formally stated in the following theorem; in particular, the result of the local power analysis is reported in part (b).

Theorem 4 Fix $\ell > 0$, and suppose Assumptions D_a(ℓ), G, and T_γ hold. Then the following statements are true.

- a. For $0 \leq \delta < \frac{1}{2}$,

$$P[|\sqrt{n}\hat{\rho}_{n,k}(\ell)| > c] \rightarrow 1$$

as $n \rightarrow \infty$, where c is any positive constant.

- b. For $\delta = \frac{1}{2}$,

$$\sqrt{n}\hat{\rho}_{n,k}(\ell) = \pi_{k,\ell} + U + o_p(1)$$

as $n \rightarrow \infty$, where U is a zero-mean $O_p(1)$ random variable, $\pi_{k,\ell} : \frac{K_\ell}{\sigma_k^2}$, and $\sigma_k^2 : \lim_{n \rightarrow \infty} \frac{\text{Var}_0(g_{n,k,i})}{\Delta^{1-\beta\varpi}}$.

The autocorrelation test exhibits similar size and local power properties under the finite-activity jump case. This is stated in the corollary below.

Corollary 2: *The conclusions in Theorems 3 and 4 remain valid if Assumption D is replaced by Assumption D' and β is set to 0.*

4 Finite-Sample Performance

In this section, we study the finite-sample performance of $L(k)$ test introduced in the previous section.

The sample paths of the log-price process are generated from a stochastic volatility with jumps (SVJs) model given by $dX_t = bdt + \sigma_t dW_t + dX_t^d$. The stochastic volatility process (common to both DGP 1 and 2, given below) is given by $d\sigma_t^2 = \kappa(\theta - \sigma_t^2)dt + \sigma_v \sigma_t dW_t^V + J_t^V dN_t^V$ with long-run variance θ , speed of mean reversion κ , and volatility of volatilities σ_v . The increments of the two standard Brownian motions W_t and W_t^V have a contemporaneous correlation of ρ (i.e., $E(dW_t dW_t^V) = \rho dt$). Jumps in volatilities are modeled by $J_t^V dN_t^V$, where J_t^V is i.i.d. exponential with mean μ_{J_v} , and N_t^V is an independent Poisson process with intensity λ_v . The pure jump component X_t^d takes either one of the following baseline specifications:

- DGP 1: X_t^d is an infinite-activity jump process as defined in Assumption D, with $0 < \beta < 2$ and the stochastic scaling factor λ_t given by $\lambda_t = \lambda_\infty + \int_0^t \varphi(t-s) dN_s$, where $\varphi(x)$ is some kernel function and $N_t = \sum_{s \leq t} 1_{\{|\Delta X_s| > \epsilon\}}$ with ϵ set to $\epsilon = 5\sqrt{\theta}\Delta^{1/3}$.
- DGP 2: X_t^d is a finite-activity jump process as defined in Assumption D', where J_t is iid $N(\mu_J, \sigma_J^2)$ and the stochastic intensity λ_t of N_t is modeled by $\lambda_t(\omega) = \lambda_\infty + \int_0^t \varphi(t-s) dN_s(\omega)$ for some kernel function $\varphi(x)$.

The parameter values of the baseline specifications are set as follows.¹⁶ For both DGP 1 and 2: $\kappa = 4.4$, $\theta = 0.014$, $\sigma_v = 0.24$, $\rho = -0.61$, $\mu_{J_v} = 0.018$, $\lambda_v = 1.39$. The kernel function is set to $\varphi(x) = \alpha e^{-\zeta x} 1_{\{x \geq 0\}}$ with $0 \leq \alpha < \zeta = 100$ (except in Figure 4(b) where $\varphi(x) = 1_{\{0 \leq x \leq \phi\tau\}} \cos(2\pi x/\tau)$ with $\phi \geq 0$ and $\tau = 60$). For DGP 1 only: $\beta = 1.25$, $a > 0$, $\bar{\lambda} = 1$. For DGP 2 only: $\mu_J = -0.03$, $\sigma_J = 0.05$, $\bar{\lambda} = 6.3$.¹⁷ There are a few exceptions: (i) we set $\bar{\lambda} = 100$ in Panel B (upper half) of Table 1, $\bar{\lambda} = 10$ in Panels A and B (lower half) of Table 1 and in Figure 2(a) and (b), and $\bar{\lambda} = 500$ in Panels C and D (lower half) of Table 1 and in Figure 2(c) and (d) in order to ensure enough jumps under different specifications; (ii) we set $\mu_J = -0.01$ in Figure 5b in order to examine test power under small mean jump sizes; and (iii) we modify the SV and jump parameters in Figure 6 in order to vary the proportion of variations due to jumps.

The sample horizon $[0, T]$ and the discretization step Δ are set as follows. For DGP 1: $T = 1$ and $\Delta = \frac{1}{20 \times 6.5 \times 60} = \frac{1}{7800}$; for DGP 2: $T = 30 \times 252 = 7560$ and $\Delta = \frac{1}{252}$. An exception is the size experiment in Table 1 and the $L(k)$ distribution plot in Figure 2, in which T

16 The parameters are set in accordance to the empirical estimates obtained in the literature. The parameters of λ_t with exponential kernel are set according to Aït-Sahalia, Cacho-Diaz, and Laeven (2015, Table 2, first column). For the parameters of σ_t and J_t (DGP 2 only), which we cannot find in Aït-Sahalia, Cacho-Diaz, and Laeven (2015), we refer to the annualized SVSCJ model parameter estimates in Eraker (2004, Table III, last column).

17 Provided that $\alpha < \zeta$, the λ_t process is stationary with unconditional (annualized) jump intensity $\bar{\lambda} = \lambda_\infty / (1 - \frac{\alpha}{\zeta})$.

Table 1. Empirical sizes of $L(k)$, BLY, and CSS tests

T	Δ	$L(3)$	$L(6)$	BLY	CSS	$L(3)$	$L(6)$	BLY	CSS
Panel A: DGP 1, fixed Δ					Panel C: DGP 2, fixed Δ				
5	1/252	0.0476	0.0466	0.3892	0.0326	0.0192	0.0068	0.1118	0.0390
10	1/252	0.0460	0.0476	0.7010	0.0336	0.0248	0.0118	0.4854	0.0432
20	1/252	0.0546	0.0552	0.9482	0.0362	0.0400	0.0214	0.8090	0.0402
30	1/252	0.0570	0.0536	0.9924	0.0382	0.0426	0.0252	0.9038	0.0420
40	1/252	0.0520	0.0572	1.0000	0.0434	0.0552	0.0354	0.9382	0.0388
50	1/252	0.0490	0.0480	1.0000	0.0384	0.0528	0.0426	0.9630	0.0500
10/252	1/9828	0.0314	0.0302	–	0.0308	0.0332	0.0370	–	0.0036
20/252	1/9828	0.0390	0.0430	–	0.0320	0.0468	0.0472	–	0.0038
30/252	1/9828	0.0482	0.0494	–	0.0348	0.0486	0.0516	–	0.0044
60/252	1/9828	0.0510	0.0488	0.1344	0.0366	0.0510	0.0492	0.1988	0.0100
120/252	1/9828	0.0514	0.0532	0.1252	0.0338	0.0572	0.0568	0.1228	0.0188
240/252	1/9828	0.0592	0.0612	0.1898	0.0378	0.0548	0.0548	0.1570	0.0246
Panel B: DGP 1, fixed T					Panel D: DGP 2, fixed T				
1	1/390	0.0344	0.0322	0.1172	0.0310	0.0108	0.0004	0.0784	0.0284
1	1/780	0.0420	0.0430	0.1318	0.0302	0.0354	0.0260	0.1362	0.0242
1	1/1950	0.0466	0.0478	0.1382	0.0362	0.0476	0.0498	0.1578	0.0132
1	1/3900	0.0476	0.0492	0.1432	0.0328	0.0482	0.0484	0.1702	0.0078
1	1/7800	0.0536	0.0540	0.1376	0.0370	0.0518	0.0530	0.1846	0.0082
1	1/23,400	0.0580	0.0564	0.1370	0.0404	0.0586	0.0580	0.2092	0.0108
60/252	1/1638	0.0412	0.0406	0.1326	0.0306	0.0284	0.0130	0.0718	0.0312
60/252	1/3276	0.0366	0.0364	0.1608	0.0328	0.0398	0.0358	0.2222	0.0226
60/252	1/6552	0.0508	0.0494	0.1540	0.0318	0.0444	0.0484	0.2150	0.0122
60/252	1/9828	0.0510	0.0488	0.1344	0.0366	0.0510	0.0492	0.1988	0.0100
60/252	1/19,656	0.0536	0.0528	0.1190	0.0332	0.0520	0.0506	0.1614	0.0052
60/252	1/32,760	0.0536	0.0550	0.1206	0.0332	0.0486	0.0476	0.1360	0.0030

Notes: The table reports the empirical rejection rate (number of rejections out of 5000 replications) under the null hypothesis of zero jump autocorrelations. The nominal level is set to 5%. The parameter values are detailed in Section 3. DGP 1 is the SVJ model with infinite-activity jumps. DGP 2 is the SVJ model with finite-activity jumps. The sample horizon is $[0, T]$ and the discretization step is Δ . Both T and Δ are measured in years. The sets of (T, Δ) combinations that are incompatible (marginally incompatible) with the asymptotic theory of the current test are shaded in gray (light gray). BLY stands for the self-excitation test in (4.2) of Boswijk, Laeven and Yang (2018). CSS stands for the self-excitement test in Theorem 4 of Corradi, Silvapulle, and Swanson (2018). BLY statistics cannot be computed for certain (T, Δ) combinations with the selected bandwidth.

and Δ vary. From the log-return series $\Delta^n_i X$, the filtered sequence is obtained as $g_{k,i} := g_k\left(\frac{\Delta^n_i X}{u_n}\right)$ with $k = 3$ (except in Online Appendix Figure A3 where k may vary). The jump cutoff sequence takes the form $u_n = c\Delta^\varpi$ with $c = 5(\widehat{IV})^{0.5}$ (where $\widehat{IV}$ is the annualized threshold IV estimator of Mancini 2009) and $\varpi = 0.3$ (except in Online Appendix Figure A4 where c and ϖ may vary). For $L(k)$ test, we adopt the Gaussian kernel $w_b(x) = \frac{1}{\sqrt{2\pi}b} \exp(-\frac{x^2}{2b^2})$, with $b = \text{s.e.}(g_{k,i}) \times n^\xi$ with $\xi = 0.7$ (except in Online Appendix Figure A5 where ξ may vary). In all simulation experiments, the number of replications is set to 5000 and the nominal level is fixed at 5%.

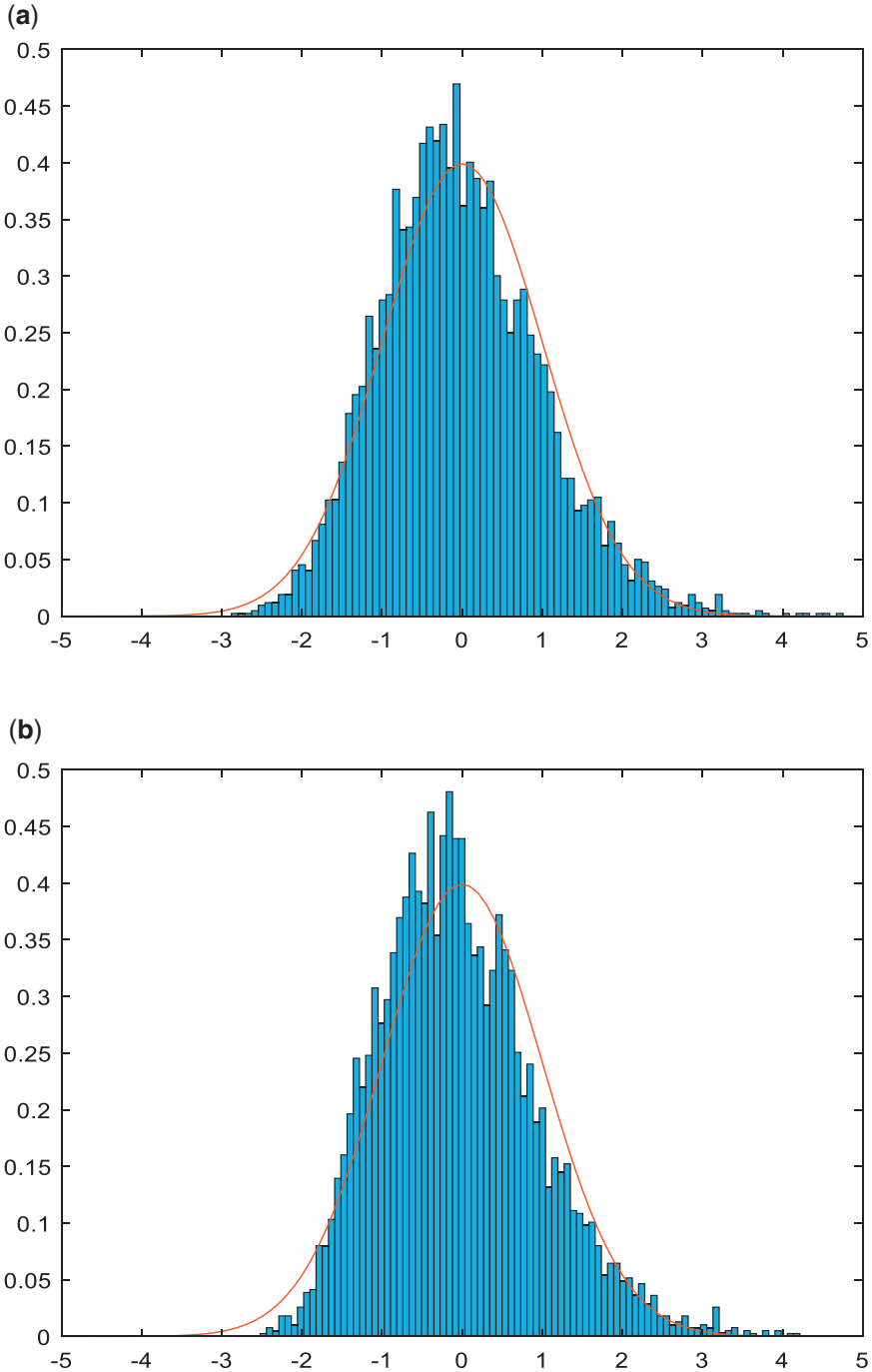


Figure 2. Empirical distribution of studentized $L(3)$ statistic in the absence of jump autocorrelation under DGP1 (infinite-activity jumps) with (a) $T = 240/252$, $\Delta = 10/(60 \times 6.5 \times 252)$ and (b) $T = 120/252$, $\Delta = 3/(60 \times 6.5 \times 252)$ and under DGP2 (finite-activity jumps) with (c) $T = 240/252$, $\Delta = 10/(60 \times 6.5 \times 252)$; and (d) $T = 120/252$, $\Delta = 3/(60 \times 6.5 \times 252)$.

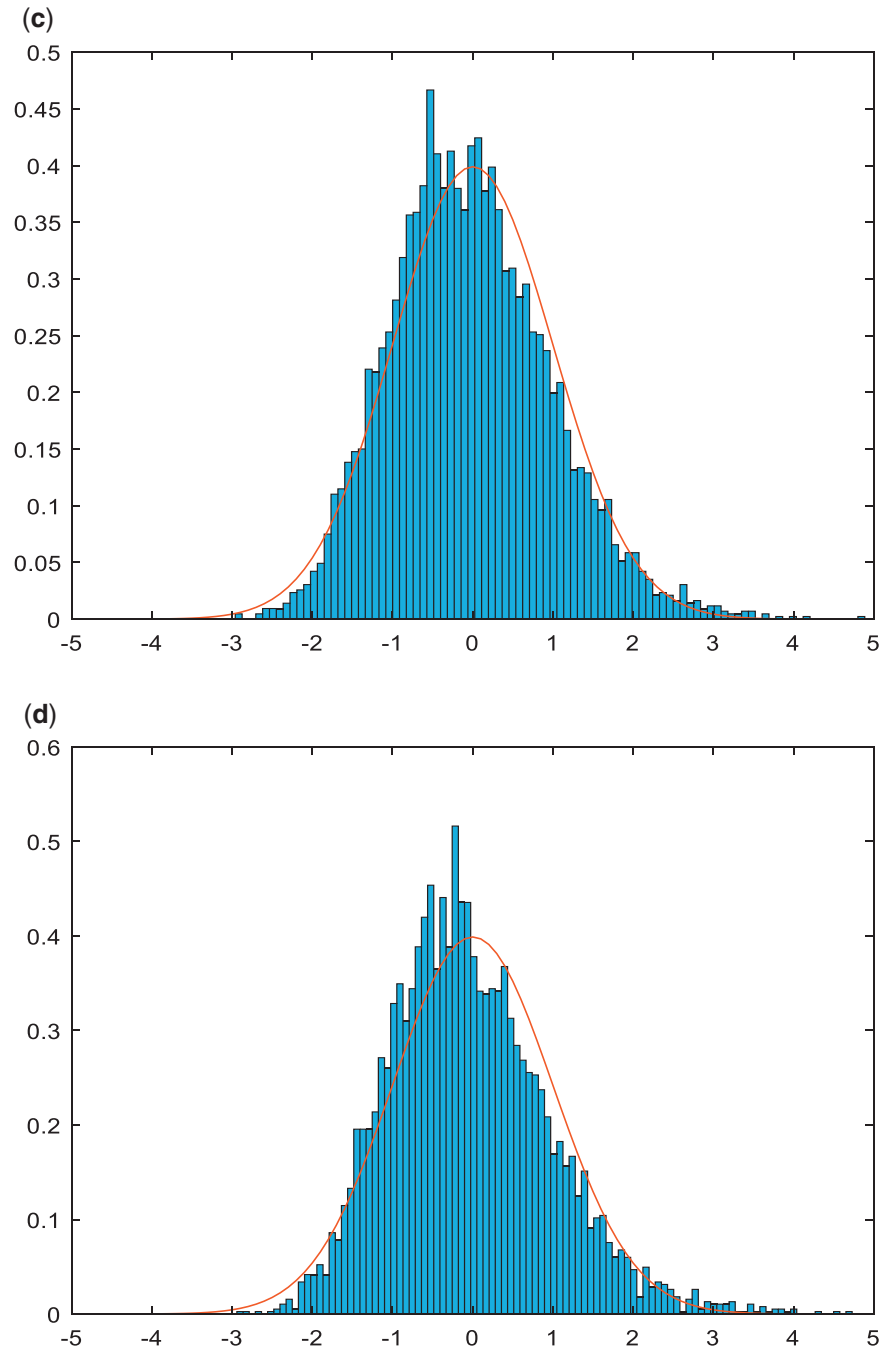


Figure 2. Continued

4.1 Size Performance

Table 1 reports the size performance of $L(k)$ test (with $k = 3$ or 6), which is compared with the competing BLY and CSS tests.¹⁸ Panels A and B present the empirical sizes when jumps are of infinite activity (DGP 1) with Δ fixed and T fixed, respectively. Panels C and D show the corresponding results under the finite-activity jump case (DGP 2). We consider a range of sampling schemes conforming to both low and high frequency data in empirical settings. The fixed- Δ scenarios are incompatible with the asymptotic theory of $L(k)$ test—Assumptions T_L and T_γ (which require $\nu > 0$) are marginally violated. Nonetheless, we include them for practical relevance.

Among the three tests, only $L(k)$ test maintains the most stable sizes across the scenarios. The comparison in Panel B (infinite-activity jumps, fixed T) is not valid for CSS test, and that in Panel C (finite-activity jumps, fixed Δ) is not valid for BLY test (and marginally for $L(k)$ test). This is because the BLY test (respectively, CSS test) is based on an in-fill (respectively, large T) asymptotics and is applicable to DGP with infinite jumps (respectively, finite jumps) only. No inference techniques can deliver consistent test results in Panel D (T fixed and Δ shrinking), because consistent estimation of autocorrelations is impossible (see Remark 4). All comparisons that are incompatible with the asymptotic theory of the tests are shaded in gray (and in light gray for borderline incompatibility) in Table 1.

The empirical sizes of $L(k)$ test stay close to the nominal level across all valid settings (Panels A–C), although the test becomes undersized in small samples when DGP contains only finite-activity jumps (Panel C). The CSS test has very good sizes (Panel C). The BLY test suffers from a mild oversize problem (Panel B)—the empirical rejection rates remain above 10% and show no sign of convergence to the nominal level as the sample grows in size.

Figure 2 displays the empirical distribution of the studentized $L(3)$ statistic when autocorrelation in jumps are absent. The standard normal distribution offers a reasonable approximation for all the sampling schemes and DGPs under consideration. We note that the empirical distribution can become right-skewed in cases where there are insufficiently many jumps in the sample series (Online Appendix Figure A8). The skewness stems from the error of approximating the chi-square distribution by $N(0, 1)$ and the rarity of jumps.

4.2 Power Performance

Next, we examine the ability of $L(k)$ test in detecting the presence of autocorrelated jumps in finite samples. Figure 3 shows the power curves of $L(3)$ test for different sample sizes. We examine two hypothetical scenarios which, respectively, mimic the fixed T and fixed Δ cases. In the first scenario (Figure 3(a)), data are generated from DGP 1 (infinite-activity jumps) and sampled over a fixed horizon (10 days) with an increasing sampling frequency ranging from 4 min ($n = 10 \times 6.5 \times 15$) to 30 s ($n = 10 \times 6.5 \times 60 \times 2$). In the second scenario (Figure 3(b)), data are generated from DGP 2 (finite-activity jumps) and sampled daily ($\Delta = \frac{1}{252}$) over a growing horizon from 5 years (with $n = 252 \times 5$) to 40 years (with

18 The BLY test we implemented is given in (4.2) of Boswijk, Laeven and Yang (2018) under the null hypothesis of “no self-excitation” with function $H = H(6, 1)$, cutoff parameter $\varpi = 0.3$, and bandwidth $k_n = \lfloor \Delta^{-0.7} \rfloor$ (for estimating the spot jump intensity. The CSS test is the “self-excitement test” given in Equation (17) of Corradi, Silvapulle, and Swanson (2018).

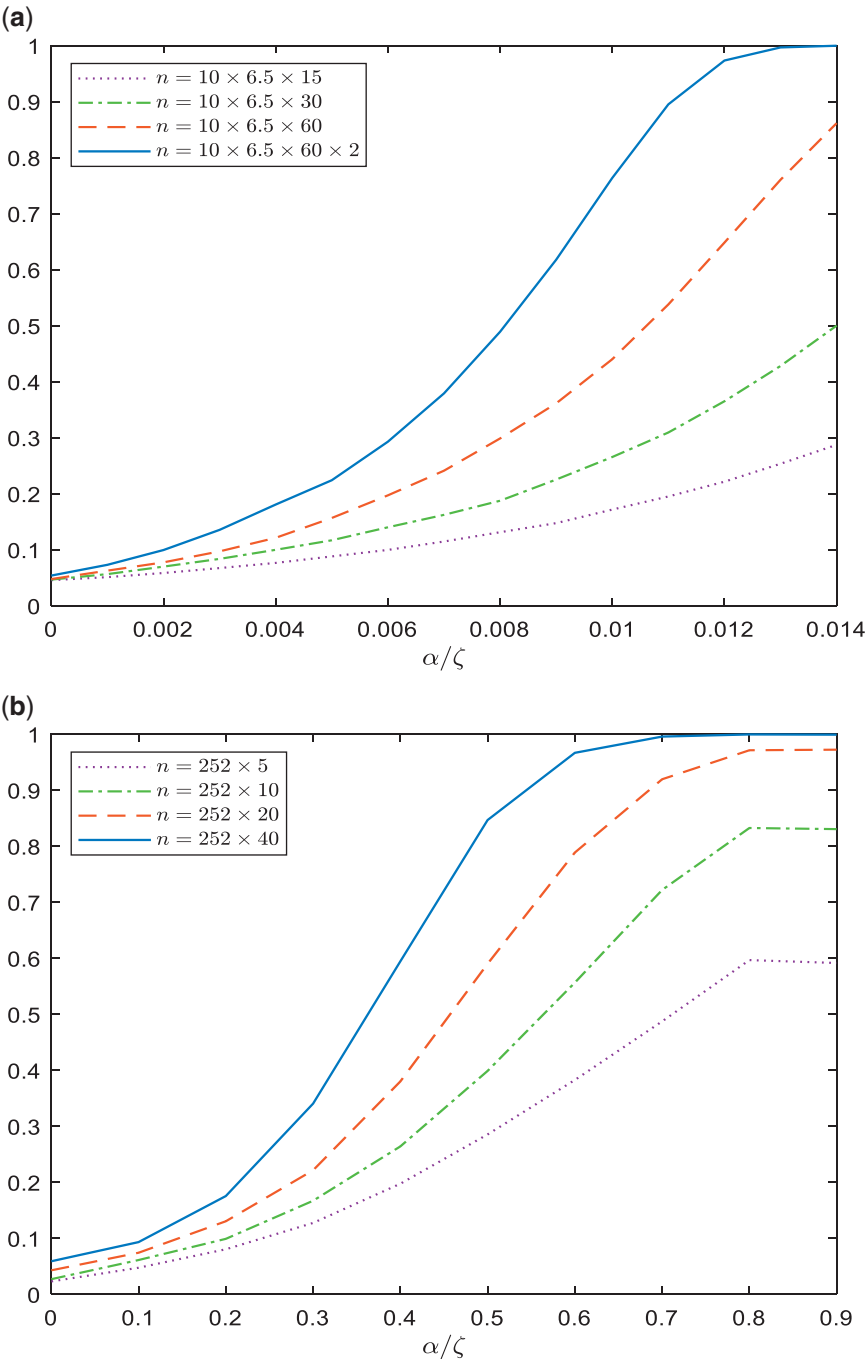


Figure 3. Power of $L(3)$ test for different sample sizes under (a) DGP 1 (infinite-activity jumps) with fixed $T = 1$ and under (b) DGP 2 (finite-activity jumps) with fixed $\Delta = 1/252$.

$n = 252 \times 40$). It is clear from the plots that, in both scenarios, the power increases with the sample size n for any given level of jump excitation.

It is of interest to compare $L(k)$ test with competing ones in the literature. In particular, the BLY test is available to detect self-excitation of jumps when they are of infinite activity. Figure 4 compares the power of $L(3)$ test with that of the BLY test (under the null of no self-excitement). The data are simulated from DGP 1 (infinite-activity jumps) with either an exponentially decaying kernel $\varphi(x) = 100e^{-\zeta x} 1_{\{x \geq 0\}}$, or a periodic kernel $\varphi(x) = 1_{\{0 \leq x \leq \phi\tau\}} \cos(\pi x/30)$. The power curves are obtained for different strength of autocorrelation, by increasing either the ratio $\frac{\alpha}{\zeta}$ for the exponential kernel (Figure 4(a)), or the number of cycles ϕ for the periodic kernel (Figure 4(b)). We keep $T = 1$ and $\Delta = \frac{1}{7800}$ throughout the simulations. We notice that the BLY test is oversized and has a lower power than $L(3)$ test as jump occurrences are more autocorrelated. Furthermore, the design of $L(k)$ test (but not the BLY test) prevents the cancelation between positive and negative autocorrelations at different lag orders. This explains the wider gap between the power curves of the two tests in Figure 4(b).

Another competing method is the CSS test designed for detecting self-excitement of finite-activity jumps. For comparison, we implement both the $L(3)$ and CSS tests on data generated from DGP 2. The power curves are plotted in Figure 5. The only difference between the subplots is in the mean jump size: $\mu_j = -0.03$ in Figure 5(a) and $\mu_j = -0.01$ in Figure 5(b). In both cases, the plots reveal that $L(3)$ test outperforms the CSS test by a large extent as the jumps are self-excited (except under weak self-excitement). When the jump size distribution is more symmetrical about zero (Figure 5(b)), the CSS test is essentially powerless.¹⁹

4.3 Robustness Analysis

In this section, we study the effects on test performance of different DGP specifications and different choice of tuning parameters.

The simulation study for the baseline specifications presumes a fixed amount of jumps relative to diffusive variation. We next study the effect on the test power due to a change in the relative contribution of jump and diffusion to log-return variation. We confine to DGP 2 (finite-activity jumps) in which the variance of jump increments is finite. Let p_J denote the proportion of log-return variance due to jumps.²⁰ The model parameter values of four model specifications with different p_J are recorded in Online Appendix Table A1. Online Appendix Figure A1 displays the power curves of $L(3)$ associated with these four model specifications. For a given level of autocorrelation, the test becomes more powerful as jumps contribute more to the log-return variation relative to diffusion.

19 This limitation is recognized and discussed in CSS (2018) and becomes the motivation for them to propose an alternative test (denoted $\tilde{S}_{t^*, \Delta}^\beta$ in their paper) based on the squared return autocorrelation of lag order one. However, when implementing this alternative test in our simulation study we find that it suffers from a serious oversize problem (which CSS (2018) confirms in their Monte Carlo findings); hence, we do not report the results of the alternative test.

20 The proportion p_J of log-return variance due to jumps, computed under the null hypothesis of no jump autocorrelations, is given by $p_J = \frac{V^d}{V^c + V^d}$, where $V^c = \theta$ and $V^d = \bar{\lambda}(\mu_J^2 + \sigma_J^2)$ represent the annualized long-run variances of the diffusion and jump components, respectively.

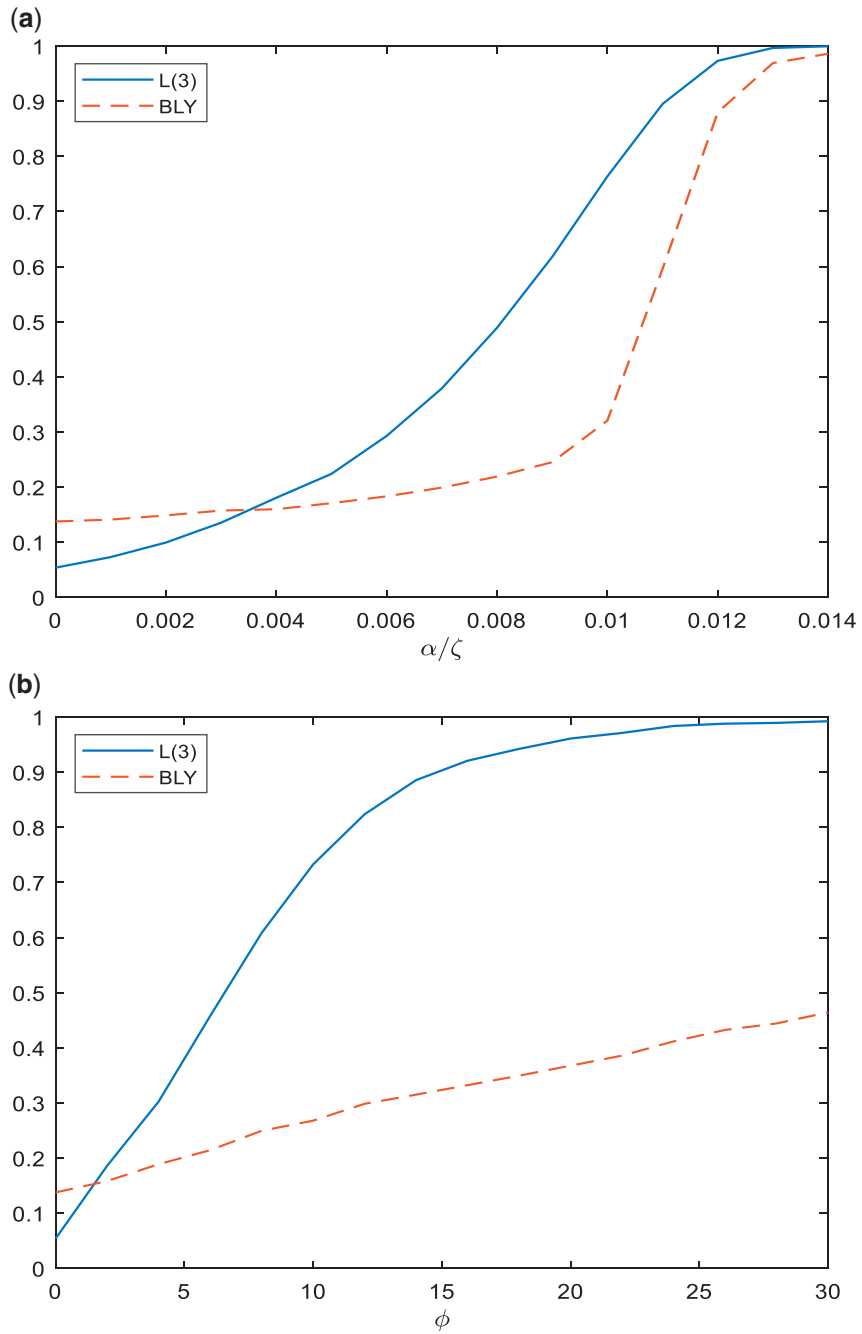


Figure 4. Power of $L(3)$ and BLY tests under DGP 1 (infinite-activity jumps) with (a) exponentially decaying and (b) periodic kernel for λ_t .

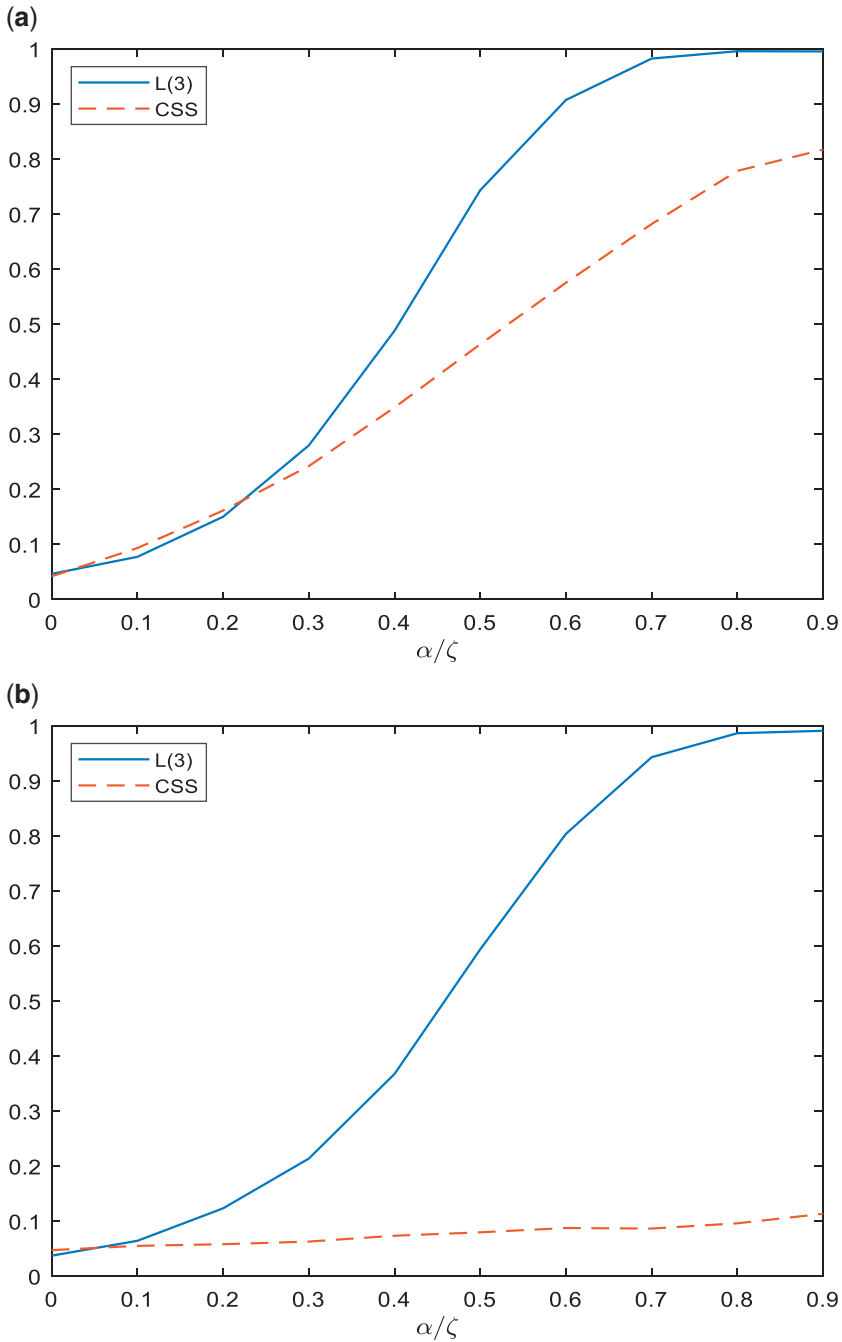


Figure 5. Power of $L(3)$ and CSS tests under DGP 2 (finite-activity jumps), with mean jump size equal to (a) $\mu_J = -0.03$ and (b) $\mu_J = -0.01$.

Somewhat related to the above study, we examine whether the test power reacts to a change in the absolute amount of jumps in the data. Since the validity of $L(k)$ test is conditional on the presence of jumps in the DGP, we expect that the test may break down when jumps are too scarce. To this end, we perform a simulation study on the test under DGP 2 with $T = 30$, $\Delta = \frac{1}{252}$, and varying unconditional jump intensities $\bar{\lambda}$, ranging from 1 to 20. This means that the data series generated from these model specifications will have an average number of jumps ranging from 30 to 600. The power curves are displayed in [Online Appendix Figure A2](#). We observe that the test power is quite invariant to the jump intensity. As expected, the test becomes less powerful when jumps are rare ($0 < \bar{\lambda} \leq 1$), and does not yield false positive results when no jumps occur ($\bar{\lambda} = 0$).

Recall that $L(k)$ test involves a number of tuning parameters. They include the power k (which determines the discount on small jumps), weight function parameter ξ (which controls the divergence rate of the bandwidth h , and hence the decay rate of the weight function), and parameters c and ϖ for the cut-off sequence.²¹ We will investigate how a change of these parameters affects the power of the test.

Starting with the robustness on the choice of k , [Online Appendix Figure A3](#) displays the power curves of $L(k)$ test for different values of k under DGP 1 (infinitely activity jumps with $\beta = 1.25$, [Online Appendix Figure A3\(a\)](#)) and DGP 2 (finite-activity jumps with $\beta = 0$, [Online Appendix Figure A3\(b\)](#)). The empirical rejection rates are plotted against the ratio $\frac{\xi}{\beta}$, which measures the extent of autocorrelation of jump occurrences in the DGP. It is clear that the power decreases with k . Recall that having larger k diminishes smaller jumps even more and leads to smaller effective sample size for jumps. On the other hand, we observe that $L(k)$ test is oversized under DGP 2 when k is smaller than three. This is because the choice of k violates Assumption G, which requires that $k > \frac{1-\beta\varpi}{1-2\varpi} = \frac{1}{1-2(0.3)} = 2.5$ under DGP 2. The requirement on k is less stringent when the jump activity level is higher; for example, for $\beta = 1.25$ we require $k > \frac{1-1.25(0.3)}{1-2(0.3)} = 1.56$. Without precise knowledge of the nature of jump activity, we recommend the user to set $k = 3$ so as to maximize the test power without violating Assumption G.

Suppose the data are commonly generated from the baseline specification of DGP 1 (infinite-activity jumps). We first conduct $L(3)$ test with different cutoff sequence of the form $u_n = c\sqrt{\widehat{IV}\Delta^\varpi}$ by changing c and ϖ . To compute the integrated volatility estimator $\widehat{IV}$, we use the nonparametric threshold estimator of [Mancini \(2009\)](#).²² The power curves ([Online Appendix Figure A5](#)) reveal that the test power tends to be higher with a smaller c and/or a larger ϖ . Intuitively, based on the definition of the filtered sequence $g_{n,k,i}$, there are more jumps to be fed into the test statistic if the sequence u_n vanishes to zero at a faster rate. However, this may come with the risk of misclassifying the larger diffusive increments as

- 21 Strictly speaking, the tuning parameters also include the form of the filter function $g(\cdot)$, the weight function $w(\cdot)$, the cutoff sequence as a function of Δ , and h as a function of n . Given what we learned from the nonparametric inference, we expect that the test result is less sensitive to the choice of the functional form than to the choice of the bandwidth parameters (c , ϖ , and ξ) that control the decay rate of the cutoff sequence and the weight function.
- 22 We may employ any other consistent integrated volatility estimator that is robust to autocorrelated jumps, for example, the threshold bipower variation of [Corsi, Pirino, and Reno \(2010\)](#). The realized bipower variation of [Barndorff-Nielsen and Shephard \(2004\)](#) tends to overestimate the integrated volatility in the presence of self-excited jumps and hence should not be used.

jumps. The misclassification could result in an oversized test when the stochastic volatility process is serially correlated. Since the empirical sizes are close to the nominal level, there is no evidence of misclassification for the choice of cutoff sequences considered in [Online Appendix Figure A4](#).

With the same DGP, we conduct $L(3)$ test with different values of ξ and obtain the power curves ([Online Appendix Figure A5](#)). It is observed that the power tends to increase with ξ for moderate level of autocorrelation, although there is no clear relationship when jump occurrences are weakly autocorrelated. This is as expected because as ξ increases the test statistic includes more lags of the autocorrelation function; however, a higher ξ reduces the relative importance on autocorrelation with smaller lags. The latter effect may lead to a less sensitive test to autocorrelation over short range.

Next, we analyze the effect of jump activity levels on the test performance. We confine to DGP 1 (infinite-activity jumps) with different values of jump activity level ($1 \leq \beta < 2$) that are plausible in high-frequency financial time series. The power curves are shown in [Online Appendix Figure A6](#). We note that the test size is insensitive to β , that the test is powerful against autocorrelated jumps with different activity levels, and that the test power increases with β for a given level of autocorrelation.

As a final simulation study, we evaluate the robustness of our test to the presence of autocorrelated jumps in volatility. We modify DGP 1 (infinite-activity jumps) by introducing volatility jumps that follow a separate Hawkes process with exponential kernel.²³ All other specifications remain the same as the baseline case (DGP 1). We want to check if the empirical rejection rate remains stable as we increase the degree of volatility jump clustering. The rejection rates are displayed in [Online Appendix Figure A7](#) for different intensities (panel (a)) and mean sizes (panel (b)) of volatility jumps. We find that the test performance is fairly robust to the presence of autocorrelated jumps in volatility. The test becomes oversized when volatility jumps occur at extremely high intensity and with strong persistence ($\bar{\lambda}_V = 1000$, $\alpha_V/\zeta_V \geq 0.7$, $\mu_{J_v} = 0.018$), and when volatility jumps occur with extremely large mean magnitude and strong persistence ($\bar{\lambda}_V = 10$, $\alpha_V/\zeta_V \geq 0.7$, $\mu_{J_v} = 0.2$).

5 Empirical Study

In this section, we apply the inference methods to study the dependence structure of jumps (if they exist) for empirical time series of stock returns. Our sample data consist of 10-min log-returns of a collection of stocks over 2006–2013. The collection ([Online Appendix Table A2](#)) contains 38 stocks that become a Dow Jones constituent at some point during the sample period.²⁴ In addition, we include SPDR S&P 500 ETF Trust (SPY) that approximates the market portfolio. The NBBO filter (see [Aït-Sahalia and Jacod 2014](#)) is employed to remove any price anomalies in the TAQ transaction data. From the NBBO-filtered data, we obtain the 10-min log-return series.

23 More precisely, the variance process σ_t^2 is assumed to have the jump component $\int_0^t J_s^V dN_s^V$, where the jump sizes J_t^V are iid exponential with mean μ_{J_v} , and N_t^V is a Hawkes process with stochastic intensity λ_t^V given by $\lambda_t^V = \bar{\lambda}_V + \int_0^t \alpha_V e^{-\zeta_V s} dN_s^V$.

24 We do not include in our sample Visa Inc. (V), which was listed as an IPO on March 19, 2008 and became a constituent stock on September 23, 2013. See [Online Appendix Table A2](#) and the footnotes for further details of our sample.

We first study the jump dynamics for each sample series over the entire sample period. For each sample series, we compute the sample autocorrelation $\hat{\rho}_{n,k}(\ell)$ (for lags $\ell = 1, \dots, 5$), and conduct $L(3)$ test and the BLY test. The results are reported in Table 2.²⁵ All except one stock have significant $\hat{\rho}_{n,k}(\ell)$ for some lags up to the fifth order (shaded in red in Table 2). The temporal dependence is predominantly in the form of jump clustering (as all $\hat{\rho}_{n,k}(\ell)$ are positive), although the magnitude and persistence of autocorrelation vary across different sample series. Furthermore, 29 out of 39 sampled stocks are detected with significant autocorrelation jointly over all nonzero lags. It turns out that the strongest rejections by $L(k)$ test occur on those stocks that were hit hard during the global financial crisis in 2008 (e.g., AIG, BAC, C, GM, and GS). By comparison, the BLY test implemented on our sample yields a rejection on merely four stocks: AIG, AXP, GM, and TRV.²⁶

Recall that the pointwise and omnibus tests are different both in the set of targeted alternatives and in their power performance. Having significant autocorrelation for specific lag orders does not guarantee a significant result when tested jointly (e.g., IBM, INTL, and MSFT). We also observe that similar estimates of $\hat{\rho}_{n,k}(\ell)$ do not readily translate to similar level of rejection by $L(k)$ test (e.g., compare BAC and UNH).

The temporal dependence of jump occurrences for four selected stocks (AIG, C, HON, and MSFT) is represented visually in Figure 6. The selected stocks exhibit a rich variety of jump autocorrelogram profiles that differ in level and shape. The jump dependence in the log-return series of AIG and C tends to be very persistent, while that for the HON series is more short-lived but stronger in magnitude. Although jumps occur in the MSFT series, their occurrences are not temporally correlated over time. Also displayed in Figure 6 are the log-returns series (the cutoff $\pm u_n$ in red dashed line), the filtered jump series $g_{n,3,i}$, and the spot jump intensity (computed according to equation (3.6) of BLY (2018), with $k_n = \lfloor \Delta_n^{-0.7} \rfloor$). The plots of $g_{n,3,i}$ show heavy clustering of jumps for AIG and C.²⁷ The jumps are clustered much more lightly (but still visible, around late 2008) for HON, while those for MSFT appear to scatter randomly without forming clear clusters. These observations conform to the result of our tests which take $g_{n,3,i}$ as input ($L(3)$ test yields rejections for AIG, C, and HON). By contrast, the BLY test rejects the null of no self-excitement of jumps only for AIG among the four stocks. The BLY test is designed to measure the co-movements between log-price and spot jump intensity, which appears to be quite strong for AIG as seen from the plots.

- 25 The significance level of all tests is set at 5%. To implement our inference procedure, we fix the end time T to 1 so that the discretization step is $\Delta_n = \frac{1}{n}$, where n is the number of observations ($n \approx 80,000$ on average). The cutoff of the filtered jump series is set to $u_n := 7\sqrt{\widehat{IV}_n \Delta_n^{0.3}}$. The Gaussian kernel is used as the weight function with bandwidth $h = n^{0.7} \text{s.e.}(g_{n,k,i})$.
- 26 We implement the BLY test with the null hypothesis of no self-exciting jumps (Section 4.1 of BLY 2018). As suggested in their paper, we adopt the function $H(6, 1)$ and tuning parameter $k_n = \lfloor \Delta_n^{-0.7} \rfloor$ in the test. Other settings are the same as those for our methods: $T = 1$, $\Delta_n = \frac{1}{n}$, same cutoff u_n and same filtered series $g_{n,k,i}$.
- 27 One may suspect whether the jump occurrences of AIG and C are too many and their autocorrelations are spuriously strong. With a higher cutoff $u_n = 11\sqrt{\widehat{IV}_n \Delta_n^{0.3}}$, the jumps of AIG and C remain quite prevalent (but fewer), and their occurrences remain highly persistent although less so compared with before; see Online Appendix Figure A9. Another potential issue is the nonstationarity in jump occurrences, which could be somewhat mitigated by a year-on-year analysis; see the next paragraph.

Table 2. Inference on autocorrelation of jump occurrences on sampled stocks

Stock	$\rho_{3,\ell}$					$L(3)$	BLY
	$\ell=1$	$\ell=2$	$\ell=3$	$\ell=4$	$\ell=5$		
A	0.04	0.10	0.02	0.03	0.02	29.4	0.8
G	0.33	0.19	0.20	0.17	0.17	6472.8	5.7
KP	0.11	0.14	0.03	0.02	0.02	40.8	2.1
A	0.03	0.07	0.01	0.01	0.00	2.0	1.8
AC	0.12	0.04	0.03	0.02	0.02	466.1	1.1
	0.19	0.11	0.09	0.08	0.04	849.0	1.2
AT	0.04	0.03	0.01	0.02	0.01	1.9	1.8
ICO	0.01	0.00	0.00	0.00	0.00	−0.2	0.9
√X	0.09	0.06	0.02	0.04	0.04	3.9	0.8
○	0.05	0.03	0.01	0.02	0.01	1.2	0.9
S	0.13	0.06	0.01	0.02	0.02	10.0	1.2
Ξ	0.11	0.03	0.02	0.02	0.02	23.0	0.9
M	0.28	0.14	0.11	0.11	0.10	991.8	−6.1
ι	0.15	0.08	0.07	0.05	0.06	166.1	1.0
D	0.05	0.10	0.03	0.06	0.02	8.5	0.7
ON	0.38	0.32	0.06	0.05	0.02	94.3	1.1
PQ	0.02	0.01	0.01	0.00	0.00	2.9	0.6
M	0.04	0.02	0.00	0.00	0.01	1.1	0.8
TC	0.02	0.01	0.00	0.01	0.00	0.1	1.1
IJ	0.12	0.06	0.04	0.02	0.01	8.5	0.5
M	0.06	0.04	0.03	0.02	0.02	23.9	1.7
πT	0.03	0.01	0.00	0.00	0.00	0.1	1.1
○	0.01	0.01	0.00	0.01	0.00	0.0	1.0
CD	0.07	0.07	0.02	0.03	0.01	7.1	0.5
MM	0.06	0.03	0.01	0.01	0.07	6.2	0.7
O	0.04	0.02	0.00	0.01	0.00	0.6	1.0
RK	0.18	0.02	0.01	0.00	0.01	15.3	0.8
SFT	0.01	0.00	0.00	0.00	0.00	−0.1	1.9
KE	0.04	0.01	0.00	0.00	0.00	0.8	1.7
E	0.07	0.03	0.01	0.01	0.01	2.4	1.1
ι	0.08	0.08	0.02	0.03	0.01	8.9	0.4
Y	0.04	0.08	0.01	0.02	0.02	1.7	0.9
	0.08	0.09	0.03	0.04	0.03	3.9	0.9
√V	0.30	0.11	0.07	0.05	0.02	88.6	2.2
NH	0.12	0.09	0.01	0.01	0.01	20.0	1.6
ΓX	0.05	0.09	0.02	0.02	0.02	5.4	0.8
Z	0.07	0.06	0.05	0.23	0.04	13.7	0.9
MT	0.04	0.02	0.01	0.02	0.01	0.5	0.8
CM	0.07	0.07	0.02	0.03	0.07	3.6	0.9

Notes: All sample series consist of 10-min log-returns over the sample period 2006–2013. Columns 2–6 contain estimates of autocorrelations of jump occurrences $\rho_{3,\ell}$ up to the fifth lag. Significant pointwise tests of $\rho_{3,\ell}$ are shaded in red. Column 7 reports the result of $L(3)$ test. Significant $L(3)$ test is shaded in orange. The last column reports the self-excitation test in (4.2) of BLY (2018). Significant BLY tests are shaded in blue. The significance level of all tests is set at 5%.

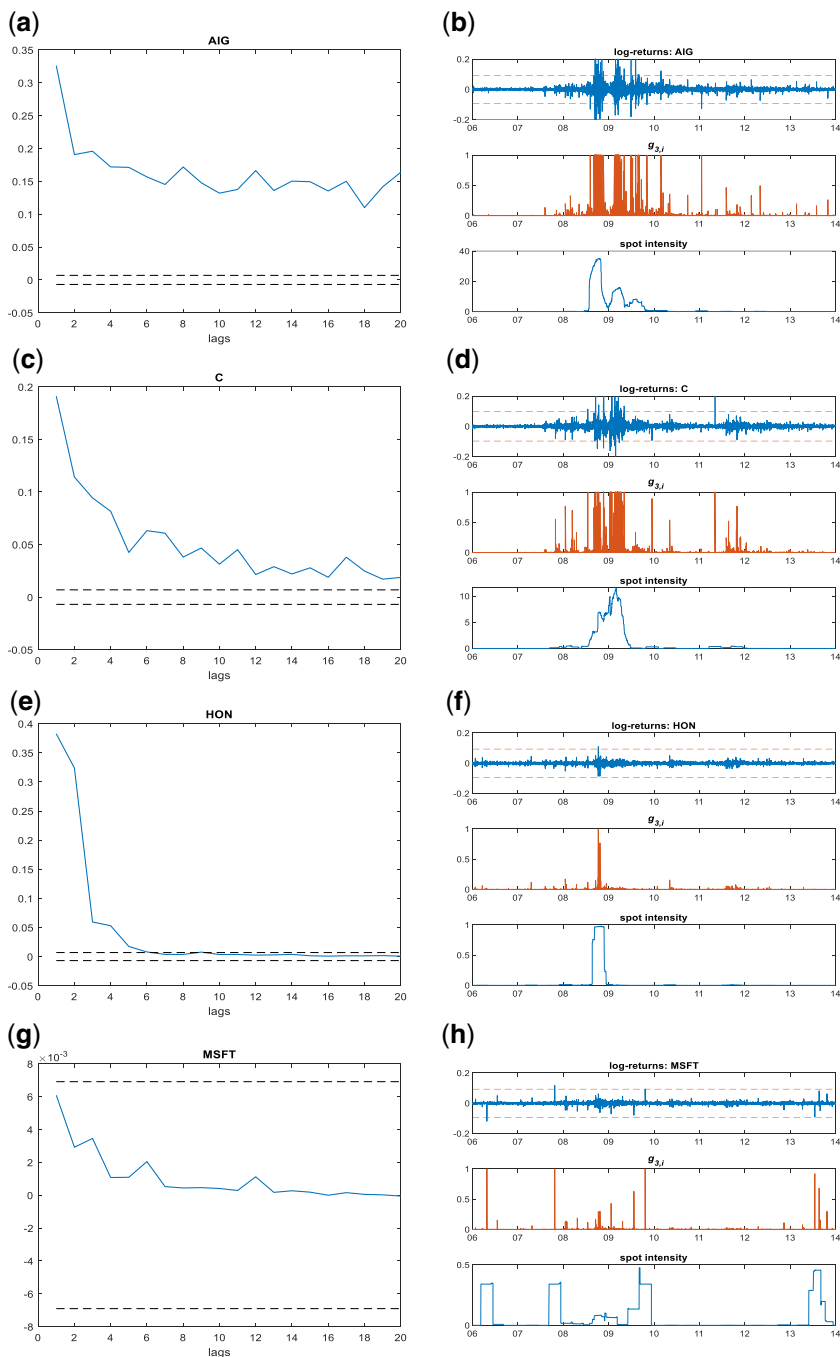


Figure 6. Autocorrelogram of filtered jump series (left) and time-series plots of log-returns (top right), filtered jump series $g_{n,3,i}$ (middle right; with cutoff $u_n = 7 \times (|V|^{0.5} \Delta_n^{0.3})$), and spot intensity of selected stocks (bottom right; computed according to Equation (3.6) of BLY (2018), with $k_n = \lfloor \Delta_n^{-0.7} \rfloor$).

The above analysis with $L(3)$ test presumes stationarity in jump dynamics over a long sample horizon spanning 8 years. Given the market fluctuations, it is quite plausible that jump stationarity is a strong assumption. Indeed, we observe from Figure 6 that the jump intensity varies substantially over time. In a similar vein, we expect that the jump dependence structure is subject to temporal variation. To investigate empirically how the degree of jump dependence may vary over time, we split the log-return series by year and then test for autocorrelation of jump occurrences. Online Appendix Table A3 reports the $L(3)$ test and BLY test results split by year. Despite a substantial reduction in sample size, $L(3)$ test yields significant results in many instances (especially in 2008), whereas the BLY test reveals only two instances (AIG and GM in 2008) of significant self-excited jumps.²⁸ Interestingly, there are clearly more stocks with significantly autocorrelated jump occurrences in 2008 than in other years. It seems to suggest a positive association between the strength of autocorrelation of jump occurrences and market downturn.

The above empirical result indicates that jumps excite themselves by different degree over cross section and over time. It is of interest to learn about the underlying attributes that drive such heterogeneity. We may get some hints by relating the autocorrelation of jump occurrences to some market variables. To this end, we consider a panel regression using the estimated lag-one $\hat{\rho}_{n,3}(1)$ as the dependent variable. The predictors include sample statistics of the high-frequency data series (average price, annual log-return, and annual volatility) and market variables that measure liquidity and size (bid-offer spread, volume, turnover, shares outstanding, and market capitalization). The regressions (Online Appendix Table A4) reveal that jumps tend to self-excite more with lower stock price denomination and more negative log-returns; in particular, it provides verification for the tendency to have clustered jumps during market downturn. Self-excitement of jumps tends to be more prevalent with wider bid-offer spread and fewer outstanding shares, both of which are associated with illiquid trading. In addition, bigger stocks (with larger market capitalization) appear to have more jumps clustered together.

6 Concluding Remarks

In this article, we introduce a set of procedures for inferring the autocorrelation of jumps as an important characterization of time-series dynamics. We develop an omnibus test for autocorrelations of jump occurrences across all lag orders under a unified analytical framework that combines in-fill and long-span asymptotics. The test delivers consistent inference under the null hypothesis of uncorrelated, stationary jumps of finite- or infinite-activity level, and is robust to the choice of sampling scheme (discretization steps and sample horizon). Being a generalized portmanteau test, it is powerful for a wide range of alternatives, including self-exciting and/or self-inhibitory jumps, jumps with long-range persistence, and jumps with cyclical autocorrelation structure. The finite-sample experiments confirm its robustness property and reveal its competitive size and power performance relative to existing tests. In addition, we advocate the use of autocorrelogram for visualizing the temporal dependence structure and conducting pointwise inference of autocorrelation up to some fixed

28 The significance level is set at 10% for all tests in Online Appendix Table A3 in light of the reduction of the number of observations ($n \approx 10,000$ on average) after splitting the sample horizon by year.

lag order. The usefulness of our inference procedure is seen in an empirical study of jump dependence dynamics on a sample of high-frequency stock return data. The inference uncovers a rich array of autocorrelation profiles of jump occurrences for different stocks over different time periods.

There are several directions of further research one may explore. As a natural extension, our procedure is adaptable to a multivariate setting for testing cross-autocorrelation of jumps from different processes. This would allow one to test for the presence of mutually exciting and/or inhibitory jumps originated from different sectors or markets. An important extension is to relax the stationarity restriction; in particular, it may allow for time-varying unconditional jump intensity. The latter can be estimated by nonparametric kernel method. One may then test for autocorrelation of “unexpected jumps” by applying the inference procedure developed in this article to the normalized residual jump occurrence process, which is covariance stationary under the null hypothesis. They are left for future research.

Supplemental Data

Supplemental data are available at <https://www.datahostingsite.com>

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